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Development and Explanation of VeriVic: An Automated Fact Checking System for Real World Claims

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1 Introduction

It is important to address and correct misinformation as falsities about true events, concepts or people can have dire consequences. For example, Andrew Wakefield’s widespread false claim that vaccines create autistic children led to make “anti-vaxxers” denying themselves and their children from receiving potentially life-saving vaccines [1]. While several fact-checking resources are available on the internet, there has been recent interest in automated fact-checking within the artificial intelligence field. Human-led fact-checking is accurate but time-consuming and arduous, especially due to the online evidence retrieval process [2]. Automated fact-checking uses transformer-based models to break down claims and produce a veracity verdict on the claim. However, it is imperative for users to have easily-readable explanations for the veracity of the claims to ensure they can be checked for accuracy.

Explanations are justifications of the claim veracity verdict and have been formatted in a variety of ways - including question and answer pairs. In this project, a system will be devised for predicting the whether claims are refuted or supported by evidence to determine their veracity. Claims will also be labelled if they have insufficient evidence or there is conflicting or cherry-picked evidence. The predictions will be justified using argumentative explanations which contain arguments derived from retrieved evidence that either supports or attack the claim. An argumentative explanation consists of a set of arguments (premises and conclusions) connected using binary support and attack relations and can be evaluated according to acceptability and circularity of arguments [3].

The result of the following research and experimentation detailed in this report culminates in the VeriVic system: an explainable automated fact-checking system for real-world claims that shows improved performance on the AVeriTeC dataset. The aims of VeriVic are outlined below:

- develop a straightforward approach to evidence retrieval
- improve the performance for predictions on the novel ‘Conflicting Evidence/Cherrypicking’ claims
- use Argumentative Explanations to improve the Averitec Score (accuracy of claim veracity predictions)

2 Background

The rapid spread of misinformation through the internet is of increasing concern, especially since false claims can quickly gain traction through social media and influence public opinion [4, 5]. The 2016 American election accentuated the threat of so-called ‘fake news’ when Facebook and Twitter users became powerful proponents of fabrications about the presidential candidates and political events, which served as political propaganda to influence voter behaviour [6]. Since the dawn of the World Wide Web in 1991, the accessibility of the Internet and its capacity for information grew to the point that the first fact-checking website, FactCheck.org, emerged in 2003. Such websites are tedious for researchers to maintain due to the scale of possible claims produced daily, and the time it takes to properly fact-check a single claim [7, 8]. As a result, automated fact-checking became a popular area of research with ClaimBuster, a tool which uses machine learning to identify political claims which require verification, being released in 2015 [9]. By 2018 the Fact Extraction and VERification (FEVER) dataset, consisting of claims annotated with evidence, was introduced alongside the FEVER shared task to develop system which determined the veracity of claims against the available evidence [10].

The FEVER task used evidence from Wikipedia articles to predict whether a claim was Supported, Refuted or had Not Enough Information [10]. However, it could not address the demand for real-world claim verification since the claims were simple and factual, unlike real-world claims, which are more complex and often contain opinions or contentious statements [2, 11]. Additionally, the evidence documents were limited to Wikipedia articles, while real-world claims require various sources due to the diversity of claims and their sub-claims [12]. Despite significant progress made by real-world claim

datasets such as PubHealth [13] and MultiFC [14], there were none which could solve the issues of claim context dependence, temporal leaking and evidence insufficiency [15]. The AVeriTeC dataset was curated to solve all the aforementioned problems with automated fact-checking as follows:

1. **Temporal leaks:** It was found that some datasets, such as MultiFC [14] had claims annotated with evidence that was released after the claim was made [16]. Fact-checking relies on past or contemporaneous information for assessment since accessing future evidence is impossible at the time of the claim. Temporal leaks were eliminated by filtering out evidence published after the claim and temporally ordering the training, development and test splits, which have different date cut-offs.
2. **Context dependence:** It was found that certain claims extracted from fact checking articles in datasets such as QABrief [17] required additional context that was present in the article in order to be verified [18]. This caused ambiguous or misleading claims that were more likely to be misclassified. AVeriTeC annotators resolved this by bolstering any such claims with the relevant information from the fact-checking article during annotation.
3. **Evidence Insufficiency:** It was found in some datasets such as Snopes [19] that labels were not based on the annotated evidence but instead based on assumptions made about the claim metadata like the speaker of the claim [16]. To avoid confusion between the aforementioned cases and actual claims without sufficient evidence, multiple stages of annotation were combined with a 'blind' control step in which claims with insufficient evidence were re-annotated.

The AVeriTeC dataset consists of claims annotated with evidence extracted from internet sources in the form of question-answer pairs. To reflect the complexity of verifying real claims where there may be both supporting and refuting evidence, a novel class was introduced for claims that had conflicting evidence or were based on cherry-picked evidence. The Conflicting Evidence/Cherrypicking claim allows systems to improve their performance on more complex claims which are not straightforward to classify. This novel class achieved the lowest performance in terms of F1 Score due to its difficulty to determine; even the AVeriTeC annotators had to re-annotate 49% of the claims due to in-annotator disagreement (figure 7).

2.1 Explainability

Explainable Artificial Intelligence (XAI) concerns designing AI systems such that their method and predictions are easily interpretable to users. It is vital for building trust that there is transparency in decision-making processes so they can be examined for accuracy and fairness and held accountable when this is not the case [20, 21]. Explainable systems allow users to evaluate if there are any errors with the verification of claims, which is especially important in the context of medical and health claims where incorrect information could lead to dangerous practices [22].

Model-agnostic explanations such as Local Interpretable Model-Agnostic Explanations (LIME) are developed after the decisions have been made by highlighting the most important words in the evidence that led to the prediction [23]. Alternatively, rules-based explanations can be embedded within the system to describe the logic of veracity predictions, usually using templates [24, 25]. Argumentative explanations use evidence to build arguments structured similarly to how a human would.

According to [3], an argumentative explanation for a model prediction is defined as a 3-tuple consisting of a set of arguments, including binary attack and support relations, the prediction and elements of the input. Such an explanation depends on Bipolar Argumentation Framework (BFA) [26] in which conflict and agreement between arguments is shown through both support and attack relations. The BFA is an extension of Dung's 1995 Classical Framework for Argumentation [27] in which an argument could have a binary attack relation with another argument to undermine its acceptability.

The following notation taken from [3] means that $\langle A, R_{Sup}, R_{Att} \rangle$ is the argumentative explanation for a prediction $\hat{y}$ using a model m and an input $\mathbf{X}$: $\langle A, R_{Sup}, R_{Att} \rangle \rightarrow_a [m(\mathbf{X}) = \hat{y}]$. R_{Sup} and R_{Att} represent support and attack binary relations between arguments respectively. For a set of argu-

ments A , $(a_i, a_j) \in R_{Att}$ means that a_i attacks a_j where $a_i, a_j \in A$. Argumentative explanations are particularly useful for Conflicting Evidence/Cherry-picking claims since their supporting and refuting evidence translates well to arguments with attack and support relations.

3 Literature Review

This chapter aims to explore and analyse the existing literature surrounding Automated Fact-checking and Explainability to identify the areas for future development that I will address in this project.

3.1 Datasets

The type of dataset chosen for veracity systems dictates how transferable they are in different domains. While there are many successful benchmarks in single domains such as health (PubHealth [13], HealthVer [28]) and politics (PolitiHop [29]), there are only two significant multi-domain datasets with real-world claims. The first being Multi-FC [14]; it lacks explanations for predictions which are vital for interpreting the prediction labels which are very many and very subject. These labels are: True, Mostly True, Half True, Mostly False, False, Mixture and Pants on Fire.

3.2 Methodologies for Automated Fact-Checking

3.2.1 Question generation

Synthetic claims are often generated using certain rules which allow them to be easily verified. For example "Donald Trump was elected president in 2016" can be fact checked using evidence retrieved from singular Wikipedia webpage. However, real-world claims are often taken from news articles or social media and require context to break them down into sub-claims for verification [30, 18, 31]. A claim from a Tweet by Chase Jennings in the AVeriTeC development dataset reads "Illegal drug, border crossings, and human smuggling activities have decreased in areas where barriers are deployed in the United States." [15]. Despite this claim being allocated 5 questions for ascertaining its veracity, it was deemed that there was still not enough evidence available to support or refute the claim. It is imperative that questions are generated which can compete with human-generated questions in terms of reasoning and thorough evidence search.

Setty et al. [32] investigated into whether it is possible to automate questions generation from claims for fact checking and compared their findings with human-generated questions. They designed a one-shot prompt with chain of thought (CoT) reasoning and used nucleus sampling to generate unique questions for a claim.

In the paper by Haoran Wang et. al (2023) [33], explanations are generated through a knowledge-grounded reasoning approach. This consists of breaking up the claim into First-order logic (FOL) clauses, retrieving knowledge-grounded answers using integrated knowledge from a search engine. The knowledge integration is useful for avoiding LLM hallucination and works well because human fact-checkers also commonly refer to Google search engines for veracity checking. However, this only is effective if the top search result is a reputable website that reports accurately.

3.2.2 Question answering task

Using self-supervised Knowledge Triplet Learning (KTL) for zero-shot question answering [34] has circumvented the necessity for question-answering annotations. This entails constructing a knowledge graph where entities and relations are represented by a triplet (h, r, t) and their embeddings are learned. Such a method relies on a structured pre-existing knowledge base such as Wikidata [35] and thus is not suitable for capturing the diversities of real-world evidence which may include conflicting evidence such as that used in the AVeriTeC dataset. Rather KTL is best suited for multiple choice answer datasets such as CommonsenseQA [36] and SocialIQA [37].

3.2.3 Evidence Retrieval

Systems like DrQA [38] and BERT-based retrievers [39] paired with BM25 [40] have been used to find relevant documents or passages that can be used to verify the claim [41]. The AVeriTeC baseline system also used BM25 paired with BLOOM-7B [42] for ranking evidence documents. In the Averitec paper, the evidence was evaluated by comparing generated questions and answers with the gold evidence that accompanied the dataset. They used a threshold for their pairwise sequence similarity scoring function, below which veracity and justification scores were zero. The chosen function was based on the Hungarian algorithm (Kuhn, 1955) [43] and was implemented using METEOR (Banerjee and Lavie, 2005) [44] from NLTK (Bird et al., 2009) [44]. METEOR was deemed superior to other methods due to its compatibilities with human judgements of similarity (Fomicheva and Specia, 2019) [45]. However, the METEOR metric presents a bias towards reference sequences and therefore penalises valid but differently phrased questions and answers. The authors acknowledged this limitation, which was also present in the FEVER dataset (Thorne et al., 2018) [46], and suggest crowd-sourced human evaluation. This solution, though effective, would incur further costs to compensate the additional labour, unless it was possible to find highly-skilled volunteers. Evidence is known to be a difficult task [47] demonstrated by the fact that useful evidence was retrieved using the claim and generated questions with a rate of only 45% [15].

3.2.4 Veracity predictions

Transformer-based architectures like BERT have proven to work well with combining the claim and evidence to determine the veracity [46]. The Averitec baseline included using a finetuned BERT-large model to decide whether each question and answer pair supported, refuted or was not relevant to the claim [15]. The methodology used determined the claim as refuted if all evidence was labelled as refuted, and similarly for supportive evidence. Where there were contradictions, the claim was labelled as Conflicting Evidence/Cherrypicking, and all other claims were deemed to have insufficient evidence.

3.2.5 Producing Explanations

The Averitec baseline utilised a finetuned BART-large model which had concatenated claim and evidence as the input [15]. It was found that including the veracity predictions did not improve the performance of the model. Haoran Wang et. al (2023) [33] explain that FOL predicates aid the LLMs in recognising the specific components of the claim which require verification, allowing them to offer more focused explanations. The experiments demonstrate that using the First-Order-Logic-Guided Knowledge-Grounded (FOLK) framework, effectiveness increases with increasing the complexity of the claim due to claim decomposition. For three-hop and four-hop reasoning, there is a 12.13% and 16.6% improvement for FOL-influenced reasoning compared to chain of thought reasoning. Additionally, they found that Knowledge-grounding is more dependable than the LLM’s intrinsic knowledge.

3.3 Evaluating explanations

The Averitec evaluation metric cannot detect when model answers contain hallucinations [15]. Haoran Wang et. al (2023) [33] highlighted that the existing evaluation metrics for claim verification may not fully capture the nuances of model performance, namely explainability in addition to accuracy. They introduce metrics which evaluate coherence, relevance, and completeness of the explanations. The qualities are important for ensuring that the explanations are very well understood by humans and don’t require further interpretation. Bingsheng Yao et. al (2023) [48] worked on developing an objective evaluation of human language explanations using a new metric which examines the helpfulness of the annotation at finetuning and inference stages instead of the popular simulatability metric. This work is useful when creating new datasets and training on them.

Tal Schuster et al. (2021) [49] found that explanations which contained conflicting evidence were deemed more informative and trustworthy by human evaluators. However, the authors acknowledged the difficulty of sourcing high-quality contrastive evidence which does not introduce bias or noise into

the dataset. This difficulty is highlighted by the fact that the Conflicting Evidence/Cherrypicking class had the lowest number of claims in the Averitec dataset. The lack of equal representation in the training and development dataset caused poor performance in this class with a precision of only 10% using gold evidence. There was an attempt to increase training data using paraphrases of the claim using the same evidence, but it did not improve the performance [15].

3.4 Conclusions

Reviewing the literature has highlighted that there is a need for greater explainability within real-world multi-domain automated fact-checking systems. Where MultiFC lacks explanations for the predictions, AVeriTeC does provide rules-based explanations. However, there is an opportunity to improve the explanations using a Bipolar Argumentation Framework with the outlook to increase the performance on the novel Conflicting Evidence/Cherrypicking class. While many systems determine argument credibility either according to evidence source or based on support-attack relations, I aim to combine the two in this project.

While systems like AVeriTeC use BM25 paired with transformer models to retrieve evidence through the document expansion method [50] which can be seen as counterintuitive [15], I aim to develop a straightforward method of extracting high quality evidence from online evidence documents.

4 Transformers

4.1 Transformer architecture

A Transformer is a fundamental neural network architecture useful for tasks such as sentiment analysis, text summarisation and question answering [51]. The main aspects of the architecture include attention mechanisms, encoder-decoder structures and feed-forward networks (Figure 1).

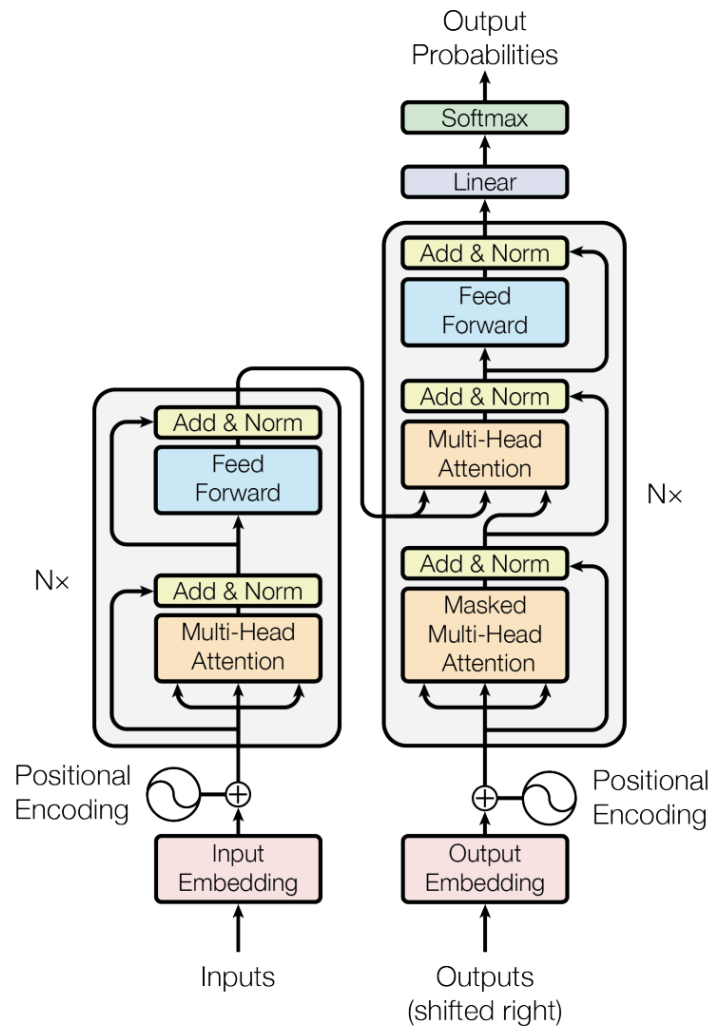


Figure 1: Transformer architecture.

Scaled Dot-Product Attention

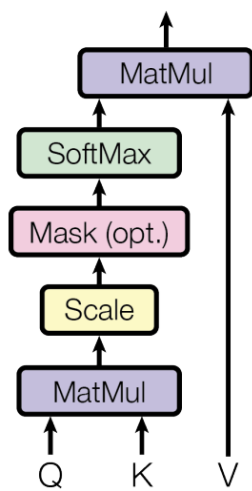


Figure 2: Self-attention. [52]

Multi-Head Attention

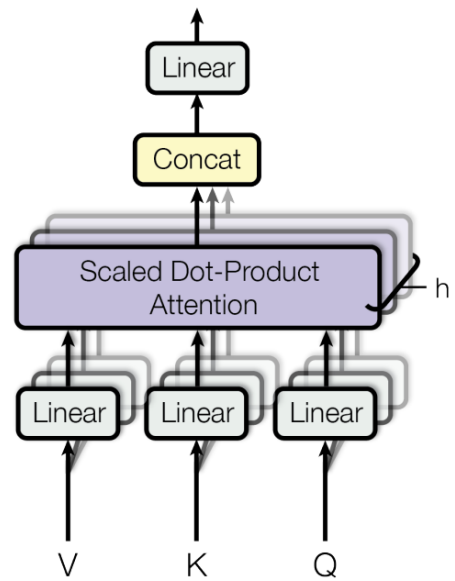


Figure 3: Multi-head attention. [52]

4.1.1 Self-attention

The self-attention mechanism (Figure 2) enables the Transformer model to consider all tokens in a sequence simultaneously instead of processing them sequentially as in a Recurrent Neural Network (RNN) [53]. While RNNs can struggle with long-range dependencies, Transformers have the ability to calculate the importance of each token relative to all others, regardless of their distance within the sequence. For example, self-attention helps to relate “cat” and “sat” in following sentence “The cat, which was very tired after a day of roaming the city, sat on the mat.”

Attention concerns vectors of queries, keys (both of dimension d_k) and values (of dimension d_v) packed into matrices Q, K, V respectively. Equation 1 depicts how the an attention score is calculated by performing the dot-products between the query with all keys and scaling it by $\frac{1}{\sqrt{d_k}}$ factor to avoid small gradients when d_k is large. Then a softmax function [54] is applied to the attention scores to convert them into a probability distribution, ensuring that they add up to 1. This allows the model to interpret the relative importance of each token in the sequence.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (1)$$

Benefits of self-attention

The parallel-structure of self-attention allows all tokens to be processed simultaneously, leading to faster training times compared to its predecessor the RNN. Since attention scores for all pairs of input tokens are computed, transformers can handle easily process sequences of different lengths without additional mechanisms. Explainability is improved because the attention scores can be interpreted to help users to understand which input tokens were most important to particular predictions. In the case of sentiment analysis, inspecting the attention scores can inform about which words contributed the most the classification. Finally, self-attention has effective scalability to large models which have been trained on large datasets, such as Generative Pre-Training 3 (GPT-3) [55].

4.1.2 Multi-head attention

When multiple self-attention heads are run in parallel, the model can capture different types of relationships between the tokens. For example, while one head focuses on semantic relationships (such as context), another may focus on syntactic relationships (such as subject-verb relations).

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (2)$$

where W_i^Q , W_i^K and W_i^V are the weight matrices which transform the input data into query, key and value vectors respectively for each attention head i.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (3)$$

where h is the number of heads (see figure 3), W^O is a weight matrix that projects the concatenated output back into the original dimension.

Benefits of multi-head attention

Since each head processes the input in a different way, the concatenation of all heads provides a more nuanced and comprehensive understanding of the sequence. While single head attention risks information loss through limited scope of processing, multiple heads with different focuses ensures that all angles are covered. Increased generalisation to unseen data and enhanced handling of ambiguity are also by-products of this. Despite having multiple head, the parallel architecture allows each head to operate independently and simultaneously, retaining efficiency.

4.1.3 Additional features

The transformer architecture consists of embeddings followed by positional encodings and encoder/decoder stacks. Instead of processing sequences in order like RNNs, positional encodings insert information about the relative position of tokens in the sequence. The encoder and decoder stacks are made up of N layers including Multi-head Attention mechanisms and Position-wise Feed Forward Neural Networks. The encoder produces a context-aware representation of the input sequence while the decoder generates the output sequence.

4.2 BERT

The Bidirectional Encoder Representations from Transformers (BERT) model is strictly encoder-based released by Google in 2018. The bidirectional encoder simultaneously looks forward and backwards at the context surrounding the token to build a deep understanding of the text and the relation between words [39].

BERT is pretrained on Masked Language Modeling (MLM) and Next Sentence Prediction (NSP) objectives. MLM mask words in a sequence at random and the model is trained to predict the missing words using the unmasked context around them. In NSP the model is trained to recognise the connection between two sentences. The pre-training objectives allow BERT models to perform well with text classification (useful for stance detection) and question answering.

BERT-large-uncased

BERT-large-uncased has 340 million parameters and does not distinguish between upper and lower case letters. For example, ‘Orange’ and ‘orange’ would be regarded as the exact same token. This allows for a smaller vocabulary size and which simplifies the model and thus accelerates training. For a task such as stance detection (determining whether a textual context supports, refutes or is neutral towards a reference text), distinguishing between cases is not usually useful.

4.3 RoBERTa

The Robustly Optimized BERT Pretraining Approach (RoBERTa) model is an improvement on Google’s BERT that was released by Meta (formerly Facebook AI) [56]. It was trained on 10× more data (160 GB) than BERT (16 GB), with a wider range of corpora. It trains using larger batch sizes for longer times than BERT which enables more stable training and improved performance on downstream tasks. Researchers removed the NSP objective as they found that it did not provide considerable improvement on BERT’s performance. They instead employed dynamic masking where different tokens are randomly masked each time the model processes the sequence, improving generalisation since each epoch produces a different training signal.

RoBERTa-large

RoBERTa-large has enhanced depth and capacity compared to RoBERTa-base due to doubling the number of encoder layers. It has also increased the number of attention heads and the hidden size by 33% leading to 2.7%× the parameters of the base version.

RoBERTa-base-SQuAD2

This model was fine-tuned on Stanford Question Answering Dataset (SQuAD2.0)[57] allowing it to achieve state-of-the-art performance on question-answering tasks. It was trained to identify an absence of answers as well as the best answer, making it suitable for real-world question-answering with AVeriTeC data where some questions do not have answers.

Table 1 shows that the parameters between BERT-large-uncased and RoBERTa-large are almost identical with the exception of the BERT model’s smaller vocabulary size due to lack of case dependency.

Parameter	BERT-large-uncased	RoBERTa-large	RoBERTa-base-SQuAD2
Number of Encoder Layers	24	24	12
Attention Heads per Layer	16	16	12
Hidden Size	1024	1024	768
FFNN Dimension	4096	4096	3072
Total Parameters	340 million	355 million	125 million
Maximum Sequence Length	512 tokens	512 tokens	512 tokens
Vocabulary Size	30,000	50,000	50,265
Dropout Rate	0.1	0.1	0.1

Table 1: Comparison of Parameters for BERT-large-uncased, RoBERTa-large, and RoBERTa-base-SQuAD2.

4.4 GPT

The Generative Pre-trained Transformer (GPT) is a LLM that was created by OpenAI in 2018 and its various iterations (GPT-2, GPT-3, GPT-4) have been widely successful in a range of NLP tasks [58]. It was pre-trained using an unsupervised learning approach with unlabelled data for a large text corpora [59, 60]. It utilises a unidirectional (left to right) decoder to generate coherent text by predicting new tokens based on the preceding tokens.

GPT-4o

Released in May 2024, Generative Pre-trained Transformer-4 Omni (GPT-4o)¹ is twice as fast and half the price of GPT-4 while providing improved performance and capabilities. This multimodal model processes and generates images and audio in addition to text. It performs well with text generation tasks under few-shot learning and even zero-shot learning if enough information is given in the prompt.

OpenAI has not publicly disclosed the model parameters for models after GPT-3 due to competitors and potential for misuse. However, they released high-level information about the model which is summarised in table 2.

Parameter	GPT-4o
Number of Layers	Not disclosed
Number of Attention Heads	Not disclosed
Hidden Size	Not disclosed
Context Window	128k tokens
Multimodal Capabilities	Text, Image, Audio
Vision Capabilities	Advanced vision processing
Audio Generation	Yes
Token Processing Speed	Up to 10 million tokens per minute
Knowledge Cutoff	October 2023

Table 2: Architecture Parameters of GPT-4o.

4.5 BART

The Bidirectional and Auto-Regressive Transformer (BART) model is used for sequence-to-sequence (seq2seq) tasks. It is trained by corrupting input sequences using arbitrary noise functions such as token masking and learning a model to reconstruct the original sequences.

¹<https://platform.openai.com/docs/models/gpt-4o>

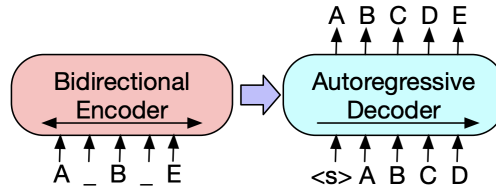


Figure 4: BART bidirectional encoder and autoregressive decoder [52]

The BART encoder-decoder mechanism (see figure 4) can be seen as a generalisation of BERT (because of the bidirectional encoder) [39] and GPT [58] (because of the autoregressive decoder). The BART bidirectional encoder processes the input sequences in forward and backwards directions to encode information from both preceding and succeeding words for a more accurate, context-aware representation. The autoregressive decoder generates the output one token at a time, ensuring each new token fits well with previously generated tokens and maintaining logical flow in the sequence as well as an entire paragraph if applicable.

BART-base model

The BART-base model is designed with 6 encoder and 6 decoder layers, each with 12 attention heads, a hidden size of 768, and a feedforward network dimension of 3072. It has a total of around 139 million parameters and can process sequences up to 1024 tokens long. It was pre-trained with a diverse corpus of large-scale English text including English Wikipedia articles and Giga5 [61] consisting of text from news sources.

4.6 T5

The Text-To-Text Transfer Transformer (T5) model is a state-of-the-art model developed by Google Research which treats every NLP task as a text-to-text problem [62]. This entails including the description of the task input and generating a text output. For example, given an input “translate from English to German: That is fantastic!”, the following output is generated “Das ist fantastisch!”.

Like BART, the T5 model is pretrained using an autoencoder objective, predicting original text after the input text is corrupted. Span masking us one of the main corruption techniques used, it entails replaced continuous spans of text with mask tokens.

T5-small model

T5 is a large model which may excessive for tasks that smaller models perform well in, such as sentiment analysis. In these cases, T5-small would be well suited for the problem as it is beneficial when computational resources such as GPU are limited. Table 3 summarises the comparison between the base and small version of T5 - notably the small model has half the number of encoder-decoder layers than its counterpart. While it enables faster training and reduces risk of overfitting, it may lead to underfitting during training.

Parameter	T5-base	T5-small
Number of Encoder Layers	12	6
Number of Decoder Layers	12	6
Attention Heads per Layer	12	8
Hidden Size	768	512
FFNN Dimension	3072	2048
Total Parameters	220 million	60 million
Maximum Sequence Length	512 tokens	512 tokens
Vocabulary Size	32,000	32,000
Dropout Rate	0.1	0.1

Table 3: Comparison of T5 Base and T5 Small model parameters

4.7 BLOOM

The BigScience Large Open-science Open-access Multilingual Language Model (BLOOM)² is a LLM created in 2022 during a collaboration with hundreds of AI researchers at a BigScience research workshop [42]. It was designed as an open-source alternative to the proprietary LLMs such as GPT. It was trained on 1.6 TB of multilingual data over many months with an emphasis on ethical AI and responsible usage.

BLOOM-7B

With only 4% of the parameters from BLOOM-176B, BLOOM-7B still performs well with text generation task but is unable to produce nuanced outputs like the larger model can. The main parameters of the BLOOM-7B model is summarised in table 4.

Parameter	BLOOM-7B
Number of Parameters	7.1 billion
Number of Layers	30
Hidden Size	4096
Number of Attention Heads	32
Attention Head Dimension	128
Feedforward Size	16,384
Context Window Size	2048 tokens
Training Data Size	1.6 TB of text
Languages Supported	46 languages

Table 4: Key Parameters of BLOOM-7B

5 Dataset Analysis

The AVeriTeC dataset consists of 4568 real-world claims, each one accompanied evidence in the form of question and answer pairs which either support, refute or are neutral towards the claim. The dataset is split in training data (3568 claims), development data (500 claims) and test data (100 claims). Each dataset split has a different cut-off date for claim publishing, with the training set having the earliest cut-off date and the testing dataset’s cut-off being the latest.

The dataset is heavily imbalanced with the Refuted claims as the majority class which represents 56.8% of claims in the training set (figure 5) and 61.0% of claims in the development set (figure 6). [15] explains that the disproportionate class distribution is due to a reliance on claims taken from fact-checking websites which prioritise fact-checking misleading or false claims. While [15] reports the highest F1 score for the Refuted class (0.69), the lowest F1 score (0.10) is achieved by the smallest class in the training set: Conflicting Evidence/Cherrypicking. During model training using imbalanced data, there is often a bias towards the majority class and deficit of meaningful learning for the minority classes which degrades performance [63].

²<https://huggingface.co/bigscience/bloom>

Claims per class - Training dataset

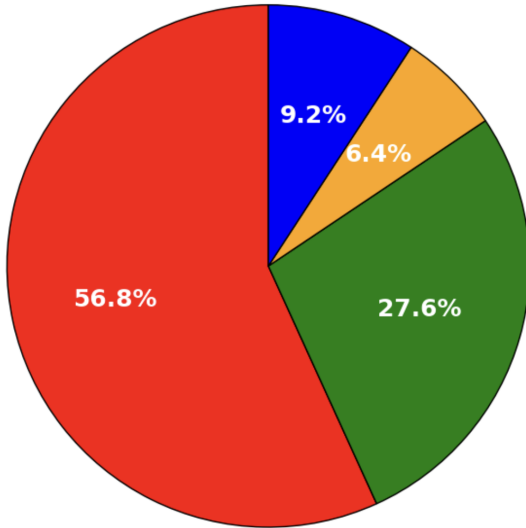


Figure 5: Percentage claims present in each label class for the training data.

Claims per class - Development dataset

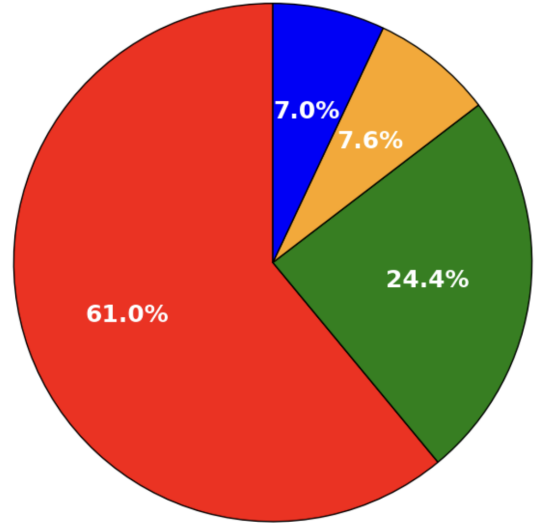


Figure 6: Percentage claims present in each label class for the development data.

Real-world claims often have both supporting and refuting evidence since events can have coverage from diverse sources which have different biases or access to different evidence. For example, a claim regarding Donald Trump’s presidency may have supportive evidence from a Republican-owned source and negative or refuting evidence from a Democrat source. It is therefore challenging to delineate between a claim that is strictly Refuted or Supported and a claim that has Conflicting Evidence/Cherrypicking. One must take source and evidence quality into consideration to filter out evidence from satirical or ‘fake news’ sources which may give low-credibility opposing evidence. This challenge is evident in figure 7 which shows that almost half of Conflicting evidence/Cherrypicking claims (49%) had to be re-labelled due to annotator disagreement.

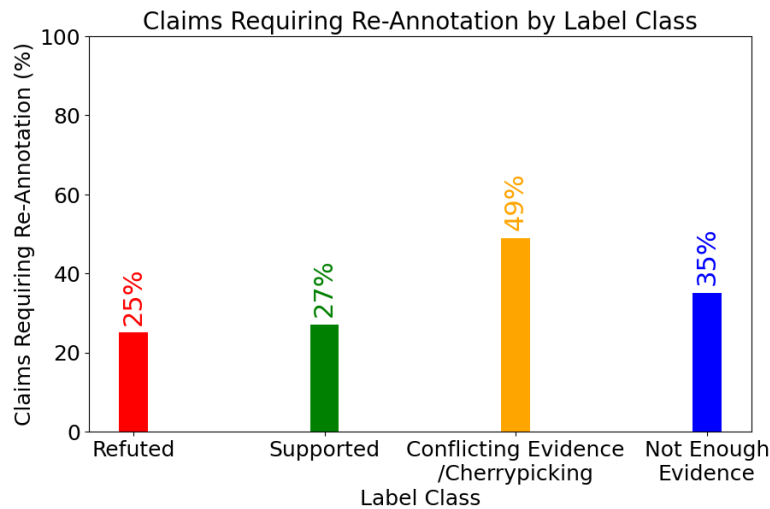


Figure 7: Percentage claims in each label class that required re-annotation during dataset curation.

Figure 8 shows the distribution of claim length by label class, which Not Enough Evidence having the

distribution peak corresponding to the highest claim length out of all classes. This indicates claims of this class tend to be longer rather than shorter. Perhaps this verbosity contributes to the difficulty with labelling (35% re-annotation rate in figure 7), since longer claims are broken down into more questions than shorter ones, each requiring evidence. While one or two questions may be answered, they depend on the other unanswerable questions to determine the claim’s veracity.

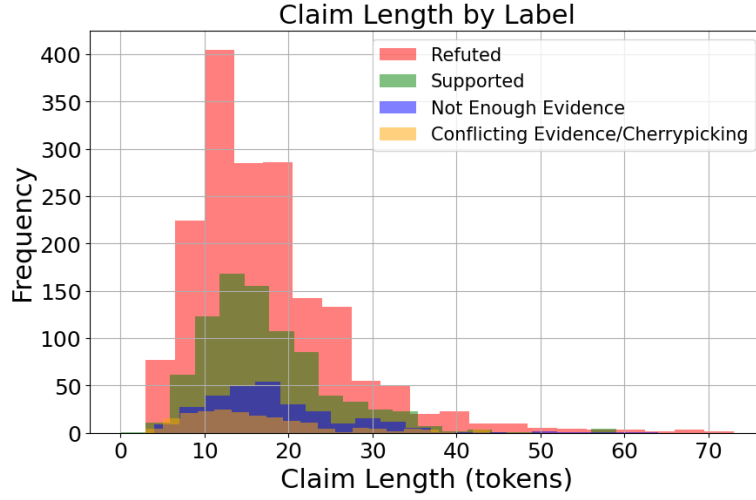


Figure 8: Distribution of claim lengths for each label class.

Figures 9 and 10 show that the majority of evidence was extracted from web text and PDF files since these sources are the most easily readable. 0.1% of evidence originated from audio clips, and while this may be simple for an annotator to access, without a transcript available, it is difficult to use this source to extract evidence. An Automatic Speech Recognition model such as Wav2Vec 2.0 by Meta would be employed in this case. 6.1% of evidence was found in the claim metadata, but several examples were not direct extractions, such as naming the speaker of the claim or its reporting source; instead, many examples were extracted from the fact-checking article from which the claim was taken.

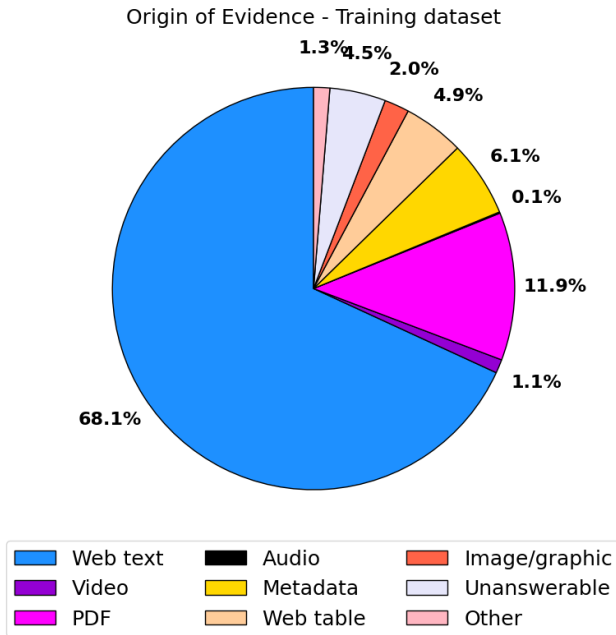


Figure 9: The percentage of claims in the training data belonging to each category.

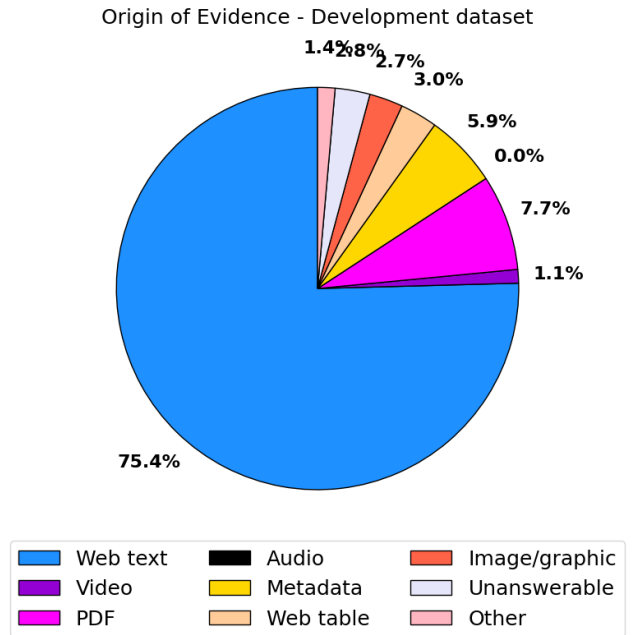


Figure 10: The percentage of claims in the development data belonging to each category.

Approximately 3/4 of the claims are Event/Property Claims or Numerical Claims for both the training

set and development set (figures 11 and 12). [15] reports that regardless of the evidence standard ($\lambda = 0.2, 0.3$), Numerical Claims had the highest accuracy, followed by Event/Property claims and Quote Verification claims. Causal Claims and Position Statements performed the worst. Despite having only the second-largest abundance in the dataset, Numerical Claims had the highest performance because numbers and dates are objective, leaving less room for subjectivity and interpretation than claims about events. It is more straightforward to cross-reference reports about specific financial figures, populations and dates than to get a consistent recount of what happened at an event.

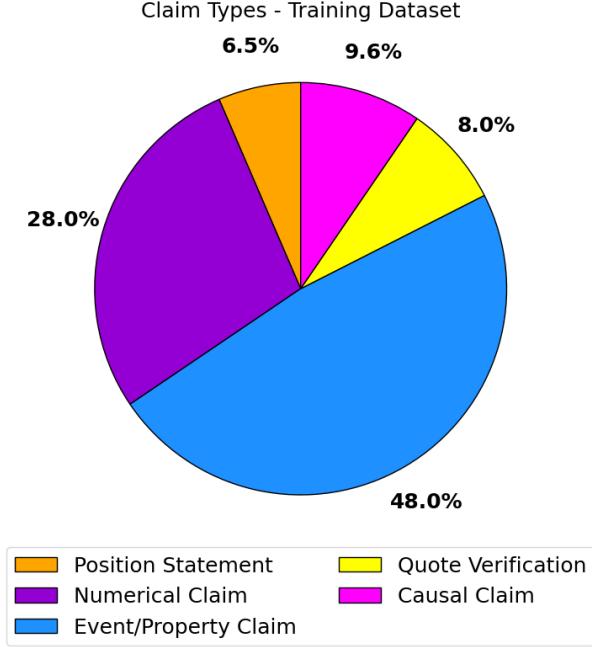


Figure 11: Different types of claim in the training dataset.

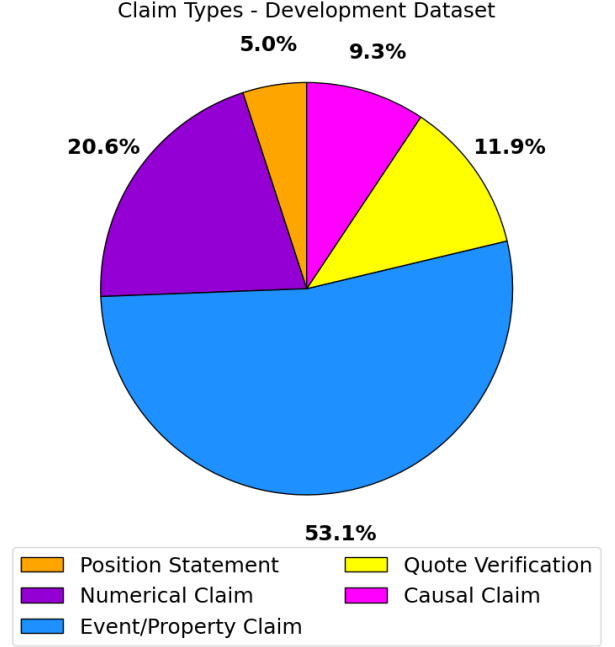


Figure 12: Different types of claim in the development dataset.

6 Evaluation metrics

6.1 Accuracy

Accuracy is used to evaluate the system’s performance on the veracity prediction task. Equation 4 shows how accuracy is calculated using two zipped lists consisting of source (predicted) labels and target (true) labels.

$$\text{Accuracy} = \frac{\sum_{i=1}^N \mathbb{I}(s_i = t_i)}{N} \quad (4)$$

where N is the total number of elements (claims) in each list, s_i is the i -th element of the source list, t_i is the i -th element of the target list, $\mathbb{I}(s_i = t_i)$ is an indicator function that returns 1 if $s_i = t_i$, and 0 otherwise.

6.2 F1

Precision

Precision (Eq. 5) is the ratio of correctly predicted positive observations (true positives) to the total predicted positives (true and false positives).

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}} \quad (5)$$

Recall

Recall (Eq. 6) is the ratio of correctly predicted positive observations (true positives) to all observations that are actually positive (true positives and false negatives).

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (6)$$

F1 score

For imbalanced datasets, such as the AVeriTeC dataset, F1 is a better metric than accuracy because it is the harmonic mean of precision and recall (see equation 7) and is unaffected by the prevalence of the majority class. High precision ensures that when a minority class such as 'Not Enough Evidence' is predicted, it's likely to be correct, while high recall ensures that as many instances of the minority class as possible are identified. Furthermore, accuracy is a misleading metric when using imbalanced data as a model could achieve a high accuracy by predicting the majority class for all instances. Doing so with the AVeriTeC development set would achieve an accuracy of 61% with zero machine learning.

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}} \quad (7)$$

Macro F1 score

The Macro F1 (Eq. 8) averages the F1 score across each class, treating every class with equal weight to ensure smaller classes are not overlooked.

$$\text{Macro } F_1 = \frac{1}{N} \sum_{i=1}^N F1_i \quad (8)$$

where N is the total number of classes and $F1_i$ is the F1 score for the i -th class.

F1 score per class

The F1 score is calculated per class to compare the performance of the model according to each class. It is expected that larger classes will have a larger F1 score than smaller classes because there was more data available for the model to learn from.

6.3 METEOR

The METEOR (Metric for Evaluation of Translation with Explicit ORdering) [64] was developed to improve on predecessor metrics' correlations with human judgements when evaluating the quality of generated text.

In this context, recall measures the fraction of words in the reference text that also appear in the generated text while precision measures the fraction of words in the generated text which also appear in the reference text. Similarly to an F1 score, METEOR utilises the harmonic mean of precision and recall, however the coefficients are changed to give a higher weighting to recall (eq. 9). The rationale for this emphasis on recall is that missing out a word from the reference text is a more significant error than adding a word to it.

METEOR allows for flexibility in the semantics of the generated text through paraphrasing, synonyms, and stemming since it is very difficult to achieve exact wording without incorporating some gold data. Synonyms are processed using the Word Net database [65] which organises words with the same meaning into sets. Stemming [66] refers to words being reduced to their base form. For example the words "ran", "runner", and "running" all have the same root: "run". Utilising these methods within the NLTK (Natural Language Tool Kit) [67] reduces penalisation for generated text which has the same meaning as the reference text but has different wording.

While the aforementioned methods help to improve the METEOR score, there is a penalisation for fragmentation which rewards texts with a word order closer to the reference text. The penalty, P , is based on how many continuous word matches there are in the generated text, where more fragmented

matches increase the penalty term and decrease the METEOR score. The METEOR is combination of the harmonic mean of recall and precision with the penalty term which measure coherence and fluency of the generated text (eq. 10).

$$F_{\text{mean}} = \frac{10 \cdot \text{Precision} \cdot \text{Recall}}{9 \cdot \text{Precision} + \text{Recall}} \quad (9)$$

$$\text{METEOR} = F_{\text{mean}} \times (1 - P) \quad (10)$$

6.4 AVeriTeC Evaluation Metrics

6.4.1 HU-METEOR (Question and Answer Pairs)

To evaluate the generated question and answer pairs, an adaptation to the METEOR, called Hungarian METEOR, was implemented. It uses the METEOR as the pairwise scoring function, f , applied to sequences in the generated sequence $\hat{Y}$ and reference text Y . Then the Hungarian Algorithm [43] is applied to decide the optimal matching of the sequences. The final scoring function, the Hungarian METEOR (HU-METEOR) [15] is given by equation 11.

$$u_f(\hat{Y}, Y) = \frac{1}{|Y|} \max \sum_{\hat{y} \in \hat{Y}} \sum_{y \in Y} f(\hat{y}, y) X(\hat{y}, y) \quad (11)$$

where $f(\hat{y}, y)$ is the METEOR function and $X(\hat{y}, y) \rightarrow 0, 1$ is a boolean function describing the optimal matching of generated and reference sequences.

6.4.2 AVeriTeC Score (Veracity Prediction) and AVeriTeC METEOR (Justification)

When evaluating the veracity predictions and prediction justifications, there is a cut of of $f(\hat{y}, y) \geq 0.25$ for deciding whether the correct evidence was retrieved in the question and answer pairs. For any claim which has an evidence score lower than 0.25, the veracity prediction and justification scores are 0.

AVeriTeC Score is the accuracy on the veracity predictions where $f(\hat{y}, y) \geq 0.25$ and the justification is scored using METEOR where $f(\hat{y}, y) \geq 0.25$.

6.5 Argumentative Explanation Metrics

6.5.1 Acceptability

Acceptability of an argumentative explanation measures whether the explanation is likely to be accepted by a human user as a valid justification for the verification decision made by the system. High acceptability is crucial in the context of explanations to encourage trust in the system's predictions.

The example argumentative explanation in figure 13 depicts argument a_1 , which is attacked by a_3 and supports a_2 , which is also attacked by a_3 . Argument a_2 is weakened by the direct attack and the attack on its supporting argument. If a_4 has a conclusion $\hat{y}$, the argumentative explanation is acceptable since a_4 has no attackers. In general, acceptability is increased if arguments a_i with conclusion $\hat{y}$ have the fewest and weakest attackers. $\text{Acc}(\mathbf{E}) \rightarrow 1$ if a_i has no attackers. Acceptability can still be relatively high if the arguments a_j which attack a_i also have many attackers.

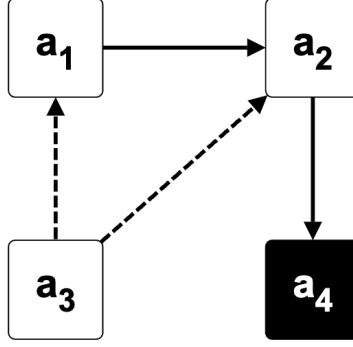


Figure 13: Arguments in an Argumentative Explanation that is acceptable because the argument a_4 with conclusion $c_i = \hat{y}$ has no attackers.

6.5.2 Dialectical circularity

Dialectical circularity measures circular reasoning which can be presented in the following ways:

- arguments attack or support themselves, i.e. the two arguments in an attack or support relation are the same
 - i.e. $(a_j, a_j) \in R_{Att}$ or $(a_j, a_j) \in R_{Sup}$
- two arguments support or attack each other
 - i.e. $(a_j, a_i), (a_i, a_j) \in R_{Att}$ or $(a_j, a_i), (a_i, a_j) \in R_{Sup}$ (figure 14)
- there is a cycle of supports involving more than two arguments
 - i.e. $(a_i, a_j), (a_j, a_k), (a_k, a_i) \in R_{Sup}$ (figure 15)

If the head and tail of an attack relation are the same argument, that argument exhibits a paradoxical scenario of contradicting itself. Similarly, an argument that supports itself may not be rooted in evidence [3, 68]. As few circular arguments as possible is ideal for an explanation, therefore it is advantageous to have a low circularity score i.e. $Cir(\mathbf{E}) \rightarrow 0$.

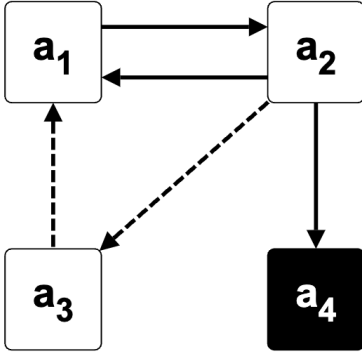


Figure 14: An example of a circular argumentative explanation due to arguments a_1 and a_2 supporting each other.

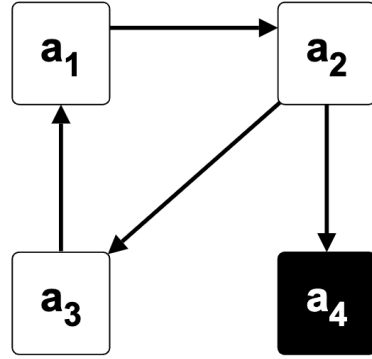


Figure 15: An example of a circular argumentative explanation due to cyclical support relations between arguments a_1 and a_2 and a_3 .

Acceptability and Circularity are formally defined by Kotonya and Toni [3] in equations 15 and 16.

7 Experiments

The AVeriTeC-BLOOM-7b system was developed to generate veracity predictions and prediction justifications for real-world claims using web-retrieved evidence. [15] highlighted the following main tasks:

search, evidence selection, veracity prediction and justification generation. Experiments were carried out in line with these tasks, however through research and as a result of the experiments, the tasks were expanded to the workflow seen in figure 16.

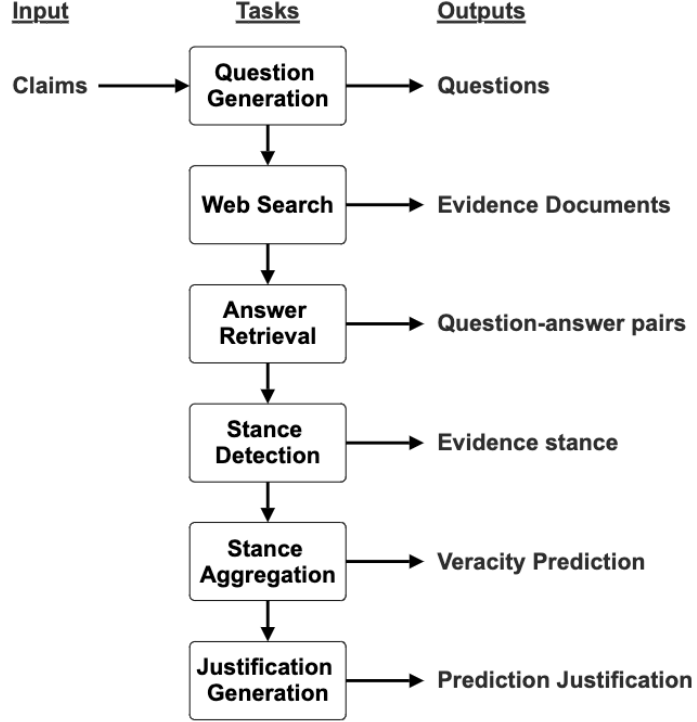


Figure 16: AVeriTeC system workflow including tasks and outputs

7.1 Question Generation

7.1.1 Direct Question Generation

The BART-base model [52] was fine-tuned using 7985 claim-question pairs derived from the AVeriTeC training data and 7552 claim-question pairs from the QuanTemp dataset. QuanTemp [69] is a real-world open-domain dataset for fact-checking numerical claims released in 2024. It was chosen for sequential fine-tuning on the BART-base model due to its similarities with the AVeriTeC dataset since numerical claims make up 28% of the AVeriTeC training data.

For the experimentation, all hyperparameters were kept the same and are summarised in table 5, with only the datasets and question generation methods changing. During inference, a total of 3 questions were generated for each claim in accordance with the AVeriTeC dataset’s average of 2.6 questions per claim. The results were written to a JSON file and evaluated using the HU-METEOR.

Hyperparameter	Value
Epochs	3
Weight Decay	0.01
Learning Rate	2e-5
Batch Size	16

Table 5: Hyperparameters used during training of BART-base model

Experiment 1

- BART-base model trained with 7985 AVeriTeC claim-question pairs
- 3 questions generated at inference on AVeriTeC development data

Experiment 2

- BART-base model trained with 7985 AVeriTeC claim-question pairs
- Pre-trained BART-base model trained again on 7552 QuanTemp claim-question pairs
- 3 questions generated at inference on AVeriTeC development data

7.1.2 FOL Question Generation

Questions can be generated by first decomposing claims into First Order Logic (FOL) predicates and converting the predicates into questions [33]. An example of this, taken from [33], is shown in table 6.

Claim: Lubabalo Kondlo won a silver medal in the 2012 SportAccord World Mind Games inaugurated in July 2011 in Beijing.	
Predicates	Questions
Won(Lubabalo Kondlo, a silver medal)	What did Lubabalo Kondlo win in the 2012 SportAccord World Mind Games?
Inaugurated(the 2012 SportAccord World Mind Games, July 2011, Beijing)	When and where was the 2012 SportAccord World Mind Games inaugurated?

Table 6: Hyperparameters used during training of BART-base model

Using the prompt in figure 39, GPT-4o generated predicates from the gold questions of 4 AVeriTeC claims, one from each label class. This created a set of 4-shot examples built from AVeriTeC data, improving claim and question style consistency.

Stratified sampling was used to compile a training dataset of 160 claims with their associated gold questions, 40 claims from each label class. The aforementioned AVeriTeC examples were used for 4-shot prompting to generate predicates from the training set gold questions. The resultant dataset of 160 claims with gold questions and generated predicates was split into 194 claim-predicate pairs and 434 predicate-question pairs.

Due to the small size of the claim-predicate pair dataset, 10 epochs were chosen for training (table 7). Conversely, the predicate-question dataset was much larger, therefore only 3 epochs were chosen for training to reduce model over-fitting.

Hyperparameter	Claim-predicate pairs	Predicate-question pairs
Dataset size	194	434
Epochs	10	3
Weight Decay	0.01	0.01
Learning Rate	2e-5	2e-5
Batch Size	2	2

Table 7: Training hyperparameters used during training of BART-base model on (1) claim-predicate pairs and (2) predicate-question pairs.

A probability distribution over the entire vocabulary for the next token was produced at every step of generating a predicate from a claim. Top-50 token sampling increased the randomness of token prediction by avoiding choosing the highest probability token each time. The top-0.95 nucleus sampling allowed the model to select from the smallest possible set of tokens whose cumulative probability was at least 0.95. Beam search was employed to simultaneously explore multiple potential sequences (number of beams = 5) to select the most likely sequence according to the model’s predictions. Early stopping was implemented to save computation time. The first model’s predicate predictions were used to generate relevant questions in the second model.

Experiment 3

- First BART-base model trained on 194 claim-predicate pairs
- Predicates generated from claims at inference
- Second BART-base model trained on 434 predicate-question pairs
- 3 questions generated from predicates at inference

Experiment 4

- Combine 3 questions from the direct (DIR) method and 3 questions from the first-order logic (FOL) method

7.1.3 Post-processing

There were some near-duplicate questions generated which were redundant and increased computation time at the search and evidence retrieval stages. Therefore the function `remove_similar_questions` removed duplicate questions whose TF-IDF vectors had a cosine similarity higher than 0.7. The result was a list of unique questions, still similar in most cases but distinct enough to retrieve slightly different evidence documents when used as queries during web search.

Experiment 5

- Remove duplicates from the questions generated in experiment 1

Experiment 6

- Remove duplicates from the questions generated in experiment 2

Experiment 7

- Remove duplicates from the questions generated in experiment 3

Experiment 8

- Remove duplicates from the questions generated in experiment 1
- Remove duplicates from the questions generated in experiment 3
- Combine these reduced lists of questions

Experiment 9

- Combine the list of questions from experiments 1 and 3
- Remove duplicates from this combined list

7.1.4 Increasing the number of questions

Certain experiments were repeated to investigate the effect of increasing the average number of questions on the HU-METEOR metrics and AVeriTeC score. These experiments are described below.

Experiment 10

- BART-base model trained with 7985 AVeriTeC claim-question pairs
- 10 questions generated at inference on AVeriTeC development data

Experiment 11

- First BART-base model trained on 194 claim-predicate pairs
- Predicates generated from claims at inference
- Second BART-base model trained on 434 predicate-question pairs

- 10 questions generated from predicates at inference

Experiment 12

- Combine 10 questions from the direct (DIR) method and 10 questions from the first-order logic (FOL) method

Experiment 13

- Remove duplicates from the questions generated in experiment 10

Experiment 14

- Remove duplicates from the questions generated in experiment 11

Experiment 15

- Remove duplicates from the questions generated in experiment 10
- Remove duplicates from the questions generated in experiment 11
- Combine these reduced lists of questions

Experiment 16

- Combine the list of questions from experiments 10 and 11
- Remove duplicates from this combined list

7.1.5 Analysis

Effect of sequential training

Table 8 shows in experiments 1 and 2 that performing sequential training on the BART-base model with the AVeriTeC and then QuanTemp data decreased the HU-METEOR score. This suggests that allowing the model to learn more about Numerical Claims does not improve the overall performance since Numerical claims already provide the highest performance out of the claim types. I hypothesise that sequential model training with a dataset more aligned with the poorer-performing claim types, such as position statements and causal claims, would improve the performance. One such dataset is the Emergent dataset [70], which contains position statements from news headlines.

Effect of DIR and FOL methods

Generating 3 questions with the FOL method (experiment 3) degraded the HU-METEOR score (compared to experiment 1). Despite efforts to increase the diversity of predicates by changing the temperature and using top-50 token sampling, there is a limit to the number of unique predicates that can be generated from a single claim. Questions generated by almost identical predicates were also almost identical. This lack of diversity for the FOL method is demonstrated by experiment 7, where questions from experiment 3 were filtered for uniqueness, leaving an average of only 1.22 questions compared to the original 3.00.

Effect of combining questions

Experiment 4 combines questions from experiments 1 and 3 to provide an average of 6.00 questions per claim. With the highest average number of questions and the highest HU-METEOR score (0.42), a clear correlation is shown. Increasing the number of questions increases the HU-METEOR score because there are more examples that can increase the similarity with the reference questions.

Effect of reducing the average number of questions

Experiments 5 to 9 demonstrate that reducing the average number of questions does reduce the HU-METEOR score. Experiment 9’s method of combining DIR and FOL questions before removing duplicates ensures that all duplicates are removed, while the inverse method in experiment 8 has a higher HU-METEOR score. While achieving the highest HU-METEOR score is beneficial, it must be considered that having duplicate questions necessarily increases the compute time for evidence retrieval without increasing the amount of useful evidence.

Exp.	Finetuned with	Method	Filtering	Avg. Q	HU-METEOR
1	AVeriTeC	DIR	None	3.00	0.37
2	AVeriTeC, QuanTemp	DIR	None	3.00	0.36
3	AVeriTeC	FOL	None	3.00	0.34
4	AVeriTeC	DIR + FOL	None	6.00	0.42
5	AVeriTeC	DIR	Remove	2.13	0.33
6	AVeriTeC, QuanTemp	DIR	Remove	1.57	0.29
7	AVeriTeC	FOL	Remove	1.22	0.25
8	AVeriTeC	DIR + FOL	Remove, Combine	3.37	0.40
9	AVeriTeC	DIR + FOL	Combine, Reduce	2.99	0.38

Table 8: Question-only HU-METEOR scores for different experiments (3 questions were generated before filtering). The ‘Exp.’ column lists the experiment number and ‘Avg. Q’ gives the average number of questions per claim.

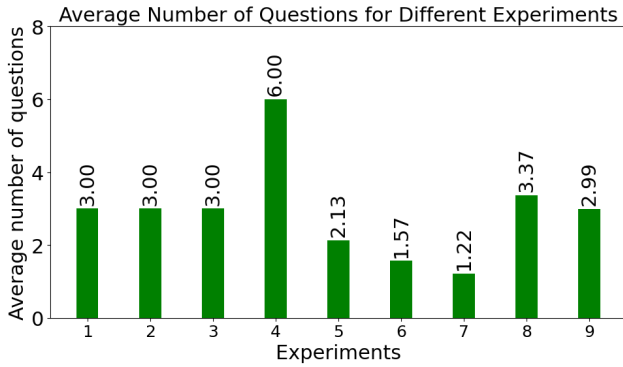


Figure 17: Average number of questions per claim for different experiments.

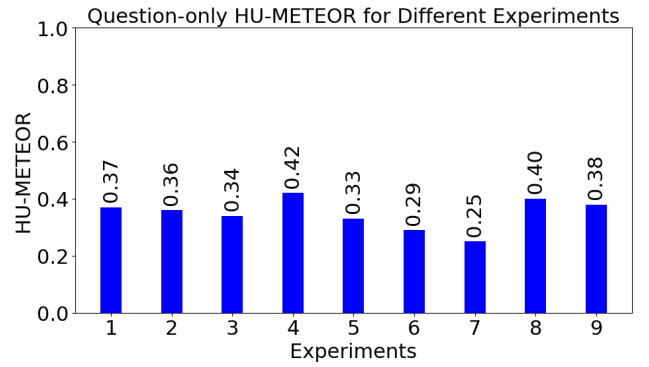


Figure 18: Question-only HU-METEOR score for different experiments.

Table 9 shows that experiment 12, with the highest average number of questions, achieved one of the lowest HU-METEOR scores due to many duplicate questions. Experiments 10 and 15 achieved the highest HU-METEOR of 0.46. However, experiment 15 was optimal since it minimised the number of questions needed to produce this score. The results of experiment 15 would be the highest performing and most efficient for downstream tasks.

Exp.	Finetuned with	Method	Filtering	Avg. Q	HU-METEOR
10	AVeriTeC	DIR	None	10.00	0.46
11	AVeriTeC	FOL	None	10.00	0.37
12	AVeriTeC	DIR + FOL	None	20.00	0.37
13	AVeriTeC	DIR	Reduce	4.65	0.42
14	AVeriTeC	FOL	Reduce	1.73	0.28
15	AVeriTeC	DIR + FOL	Reduce, Combine	6.38	0.46
16	AVeriTeC	DIR + FOL	Combine, Reduce	5.79	0.44

Table 9: Question-only HU-METEOR scores for different experiments (10 questions were generated before filtering). The ‘Exp.’ column lists the experiment number and ‘Avg. Q’ gives the average number of questions per claim.

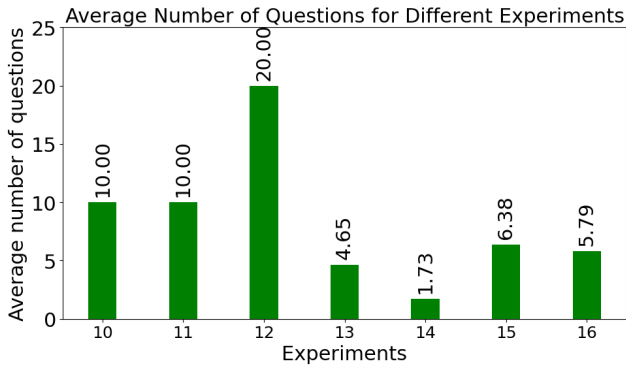


Figure 19: Average number of questions per claim for different experiments.

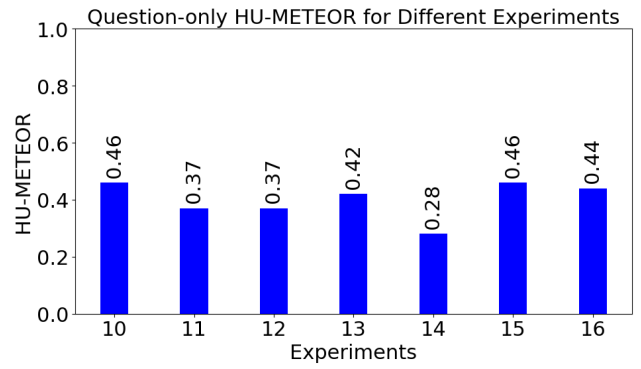


Figure 20: Question-only HU-METEOR score for different experiments.

7.2 Search

7.2.1 Metadata Evidence

Metadata gives information about the claims such as the reporting source of the claim or the claim date. Despite only 6% of evidence originating from metadata (figures 9 and 10), it is important to consider because a claim may be refuted because of some aspect of the metadata. For example, the claim “Hunter Biden, son of US President Joe Biden has died.” was refuted due to metadata evidence about the publishing date of the claim paired with information about Hunter Biden’s state of living on that date.

During experimentation with web search, interrogative queries, which asked who made the claim, when the claim was made, and what the claim was, were failing to produce relevant results. By using ‘if’ statements to check for the presence of such questions, the relevant metadata was returned as the answer, i.e. the speaker, claim date and claim text, respectively.

7.2.2 DuckDuckGo Search

Questions that asked who a person was or what an organisation was were passed through the DuckDuckGo API. DuckDuckGo is a privacy-conscious search engine which does not track or store user information. Although the API failed to return results for most question formats, it produced coherent and accurate answers when queried about named entities. For example, the question “Who is Billie Eilish?” returned a relevant, concise response taken from Billie Eilish’s Wikipedia page. However, when the same question was given to the Google Search API and the Billie Eilish Wikipedia page was retrieved, unnecessary additional text, such as web page headings, was present.

Question	DuckDuckGo answer	Google search answer
Who is Billie Eilish?	Billie Eilish Pirate Baird O’Connell is an American singer and songwriter. She first gained public attention in 2015 with her debut single "Ocean Eyes", written and produced by her brother Finneas O’Connell, with whom she collaborates on music and live shows. In 2017, she released her debut extended play, Don’t Smile at Me. Commercially successful, it reached the top 15 of record charts in numerous countries, including the US, UK, Canada, and Australia. Eilish’s first studio album, When We All Fall Asleep, Where Do We Go?, debuted atop the US Billboard 200 and UK Albums Chart. It was one of the year’s best-selling albums, buoyed by the success of its fifth single, "Bad Guy", Eilish’s first number-one on the US Billboard Hot 100. This made her the first artist born in the 21st century to release a chart-topping single.	From Wikipedia, the free encyclopedia American singer-songwriter (born 2001) Billie EilishEilish in 2023Born-Billie Eilish Pirate Baird O’Connell (2001-12-18) December 18, 2001 (age 22)Los Angeles, California, USOccupationsSingersongwriterYears active2015–presentWorks Discographysongs recordedParentMaggie Baird (mother)RelativesFinneas O’Connell (brother)Brian Baird (uncle)AwardsFull list-Musical careerGenresPopalt-popelectropoppop rockInstrument(s) Vocals guitar piano

Table 10: Comparing an answer from DuckDuckGo to an answer retrieved using Google Search.

7.2.3 Google Search Engine API Configuration

API keys

An API (Application Programming Interface) key is a unique code used as an identifier for a user to access the search engine. Google limits an API key to 1000 searches per day, several API keys were required. In total, 110 API keys were generated using the Google Custom Search JSON API, each covering approximately 21 queries. To circumvent the limit of approximately 12 API keys per user, several Google accounts were used to generate the necessary amount.

Programmable Search Engine

A Programmable Search Engine was created with the following search feature settings: ‘SafeSearch’, ‘Search the entire web’. Image search was not relevant to the AVeriTeC task, which is seq2seq, and thus was not selected. SafeSearch was selected to censor explicit content which may be retrieved from provocative queries. It must be noted that some claims in the AVeriTeC data reference sensitive topics such as paedophilia, which must be carefully processed to avoid potentially sharing offensive data. Searching the entire web was chosen because the annotators had access to the entire web during dataset curation.

Sites To Exclude

Social media websites: Social media website such as Instagram (due to low credibility) and Youtube (due to difficulty to extract information from videos) were filtered out of the search.

Known sources of misinformation: The AVeriTeC-BLOOM-7b system included a list of websites known for misinformation which were also excluded from the search. Although excluding these websites from the search may have increased the quality of the evidence documents retrieved, it may have negatively impacted the AVeriTeC score since the the annotators of the dataset had access to them while compiling the gold evidence. A second custom search engine was created which did not exclude these site, to help improve the retrieval score (question and answer HU-METEOR).

7.2.4 Google Search

For each claim, every question was used as a Google search query with a maximum of 10 results returned. Results returned for each query were filtered by publishing date to ensure that the evidence was available before the claim date and ensure that there was no temporal leaking [71].

Processing the dates

Some results contained the publishing date as part of its metadata while others had their publishing date in the result snippet (the first sentence of the web page). A function was created using regex to extract a date string from the snippet, check whether it was in a valid date format and if so convert it to a time-zone independent International Organisation for Standardisation (ISO) date format. The ISO format is given as YYYY-MM-DDTHH:MM:SS.000Z referring to year (YYYY), month (MM), day (DD) with time given in hours (HH), minutes (MM), seconds (SS), milliseconds (.000) and Coordinated Universal Time (UTC) zone. Table 11 shows an example of conversions from extracted date strings to ISO format - all time information was set to zero as it was not available from the web page.

Original date	ISO format
Nov 2, 2017	2017-11-02T00:00:00.000Z
Dec 15, 2020	2020-12-15T00:00:00.000Z

Table 11: Converting dates to ISO format.

Potential Improvements

Certain pages such as Wikipedia did not have a publishing date available but were a useful reputable source, especially for non-time sensitive information such as a description of a legal act. To address the lack of publishing data for Wikipedia pages, one could access historical revisions to pages. Using the MediaWiki API and a target date, the content of the page at that date (if available) can be inspected. This enabled the system to access only information that was available on Wikipedia before the date of the claim.

7.3 Evidence Retrieval

The AVeriTeC-BLOOM-7b system used a method inspired by the document-expansion strategy presented by [50] where initial search queries were generated to produce a set of evidence document from which sentences were ranked according to similarity to the claim. BLOOM was used to generate questions that would be answered by the top answer sentences, with the question-answer pairs re-ranked to ensure the maximum HU-METEOR score [15]. I aimed to replace this elaborate method with a straightforward solution, leveraging the questions generated in section 7.1, which already achieved a superior HU-METEOR score.

7.3.1 Evidence Sentence Ranking

The Beautiful Soup Python library³ was used to parse the HTML evidence documents to extract sentences. Sentences were coarsely filtered by checking that they included at least all of the adjectives and proper nouns in the question using Part-of-Speech (POS) tagging [72, 73] and then ranked according to how many nouns and verbs from the question they had in common. The most similar sentence from each evidence document was ranked against the others according to the number of common nouns and verbs with the question. This method was effective where relevant evidence documents were selected. Otherwise, the answers were not relevant. In the case of a claim concerning the Indian Nadar community, web pages about the French photographer Nadar populated the results, leading to incorrect answers retrieved.

7.3.2 Context Selection and Question Answering

Questions that still had no answer were subject to two question-answering models: T5-base and RoBERTa-base-SQuAD2. T5 has shown state-of-the-art performance on question answering tasks while RoBERTa-base-SQuAD2 was fine-tuned on question answering data. The context for question answering was created by splitting the top evidence document into paragraphs of 100 words and choosing the paragraph that was most similar to the question. The two models then generated an answer to each unanswered question using the context extracted from the web page.

³<https://beautiful-soup-4.readthedocs.io/en/latest/>

Potential Improvements

Many Portable Document Format (PDF) files were returned as evidence documents from the search. However, despite many attempts to rectify the issues, the function responsible for reading PDF files did not work and each PDF encountered timeout errors which were never resolved. The solution was to manually remove these results to resume evidence scraping, however this is an area for improvement in the future since 7.7% of evidence in the development set originated from PDF files.

Additionally, websites requiring the acceptance of cookies produced a timeout error and were also manually removed to resume the HTML parsing. However, in the future the requests library⁴ or the http.cookiejar module⁵ could be used to handle cookies.

7.4 Stance detection

Once the question-answer evidence pairs were generated, it was necessary to perform stance detection to determine which evidence supported, refuted and was neutral towards the claim. BERT-based models are known to perform well with stance detection tasks and were therefore the basis for experimentation.

7.4.1 Preparing training data

While the official AVeriTeC training dataset was heavily imbalanced with “Refuted” claims as the majority class, stratified sampling was used to ensure a balanced training set for the stance detection models. A dataset of 500 claims containing 125 claims from each of the four label classes was produced. According to the original paper, refuted claims only contained evidence which refuted and similarly for supported claims. While claims with not enough information contained only neutral evidence in which no answer could be found for a question or the evidence was objective in regards to the veracity of the claim. I leveraged this knowledge to reduce labelling time by forgoing manual assessment of evidence stance (refute, support, neutral) in favour of giving all evidence a single label depending on the gold claim label. Explicitly, all evidence in the refuted, supported and not enough evidence classes were labelled as ‘attack’, ‘support’ and ‘neutral’ respectively.

Conflicting Evidence/Cherrypicking claims

Conflicting or cherry-picked claims had both supporting and refuting evidence, which required a manual analysis of how each question-answer evidence pair related to the veracity of the claim. This was a time-consuming task that entailed reading through each piece of evidence and determining whether it supported or refuted the claim. Many answers were word-intensive and required reasoning to determine the stance.

One such example is shown in table 12. The first evidence took the form of a definition of eugenics which is relevant to the claim but does not indicate whether or not Sanger was a eugenicist. However, this evidence could be cross-referenced with others to support or refute the claim. The second and fourth evidence pairs both support the claim, with their answers aligning with the definition given in the first evidence pair. However, the third evidence pair attacks the claim since it describes Sanger as a progressive person who believed in racial integration (instead of discriminating against minorities).

Certain evidence answers were self-contradicting, with both supporting and attacking statements made. To tackle this, a novel label ‘conflicting’ was used to label such evidence so the model could learn to identify such contradictions. However, due to a small number of examples, the models were not able to fit this data, and consequently, no ‘conflicting’ labels were assigned during inference.

An alternative method included ‘neutralising’ this support and attacking elements of this evidence to label it neutral. However, that would have introduced an ambiguity between truly neutral evidence that did not comment on the veracity of the claim and evidence that both supports and attacks the claim. I also considered giving such evidence two labels, ‘support’ and ‘attack’. However, I

⁴<https://requests.readthedocs.io/en/latest/>

⁵<https://docs.python.org/3/library/http.cookiejar.html>

determined that having two identical texts given opposing labels would confuse the models. Both of these alternates were not investigated since they were deemed ineffective.

Claim: Margaret Sanger was a racist who believed in eugenics. Her goal when founding Planned Parenthood was to eradicate minorities. Label: Conflicting Evidence/Cherrypicking		
Question	Answer	Stance
What is eugenics?	Eugenics is the practice or advocacy of improving the human species by selectively mating people with specific desirable hereditary traits. It aims to reduce human suffering by ‘breeding out’ disease, disabilities and so-called undesirable characteristics from the human population	Neutral
What were Sangers views on eugenics?	I think the greatest sin in the world is bringing children into the world – that have disease from their parents, that have no chance in the world to be a human being practically. Delinquents, prisoners, all sorts of things just marked when they’re born. That to me is the greatest sin – that people can – can commit	Support
Was Sanger a racist?	As her biographer Ellen Chesler told me, she was a progressive who believed in racial integration. She voted for Norman Thomas. She worked with progressive Black people - W.E.B. Du Bois, for example, who along with Mary McCleod Bethune and Adam Clayton Powell Sr. served on the board of the Negro Project, a network of birth control and maternal health clinics Sanger established in Harlem and the South. In 1966, Martin Luther King accepted Planned Parenthood’s first Margaret Sanger Award, and in his statement offered a vigorous endorsement of voluntary birth control.	Attack
Was Sanger a eugenicist?	Although she did not single out Black people, Sanger was, yes, a eugenicist. She thought people, especially poor people, often had too many kids to care for properly and that too many of those kids were born physically disabled (or in the language of the day, ‘feeble-minded’). She did not oppose forced sterilization.	Support

Table 12: Stances for question and answer evidence for a ‘Conflicting Evidence/ Cherrypicking’ claim.

In the AVeriTeC baseline system, only Supported, Refuted and Not Enough Evidence claims were used for training the stance detection model while all Conflicting Evidence/Cherrypicking claims were discarded. The lack of representation in the training set may have been the reason why the F1 score for this class (0.10) was much lower than the others (0.41, 0.69, 0.16 respectively). I opted to include examples of Conflicting Evidence/Cherrypicking claims in equal proportion to the other classes to allow the finetuned models to fit for such examples.

7.4.2 Model Finetuning

Several experiments were run in order to produce the best AVeriTeC score, and they are summarised in table 13. Both BERT-large-uncased and the more robust RoBERTa-large models were finetuned using different text processing, input contexts and training-test splits to produce the best performance.

Training Parameters

Since only 500 claims were labelled for the stance detection dataset, the training split was increased from the standard 0.8 [74] to 0.9. It was found that with training epochs of 5, 7 and 10 the model began to overfit to the training data after the 3rd epoch, evidenced by the validation loss increasing after an initial decrease. Therefore, the number of training epochs was set to 3 which ensured that the validation loss did not increase. A small batch size of 8 was chosen for all experiments due to increased noise within a the small batch leading to better generalisation [75]. Furthermore, the models were variants of BERT models which were large and memory intensive - smaller batches allow for training on GPUs with limited memory [76].

Claim and Context Preprocessing (PP)

Regular expressions were used to remove contractions in the texts, for example: “don’t” became “do not”. It is a crucial step in text processing because it decreases the number of unique tokens and

simplifies the models’ task which helped to improve the performance [67, 77]. Certain contraction such as “it’s” introduce ambiguity because they can be expanded into “it has” or “it is” - removing this uncertainty helps the model better understand the text [78]. Additionally, any special characters were removed from the text while retaining any numbers. Numbers were important in the claims and evidence because dates, money and population numbers featured heavily throughout the datasets - 37% of training set examples were numerical claims.

Experiment	Model	PP	Context	Test size
1	BERT-large-uncased	No	Question, Answer	0.1
2	BERT-large-uncased	Yes	Question, Answer	0.1
3	BERT-large-uncased	Yes	Answer	0.1
4	BERT-large-uncased	Yes	Answer	0.2
5	RoBERTa-large	No	Question, Answer	0.1
6	RoBERTa-large	Yes	Answer	0.1

Table 13: Experiments conducted. PP: claim and context pre-processing. Test size: fraction of dataset set aside for evaluation of models during training.

7.5 Veracity Predictions

7.5.1 Rules-based method

Taking inspiration from the labelling process used for the AVeriTeC dataset, I devised an algorithm for predicting labels based on evidence stances which is demonstrated in figure 21. Claims with only ‘attack’ evidence stances were labeled as ‘Refuted’, claims with only ‘support’ evidence stances were labeled as ‘Supported’ and claims with only ‘neutral’ evidence stances were labeled as ‘Not Enough Evidence’. Finally, a claim was labelled as ‘Conflicting Evidence/Cherrypicking’ if it contain both ‘support’ and ‘attack’ stances or it contained a ‘conflicting’ stance.

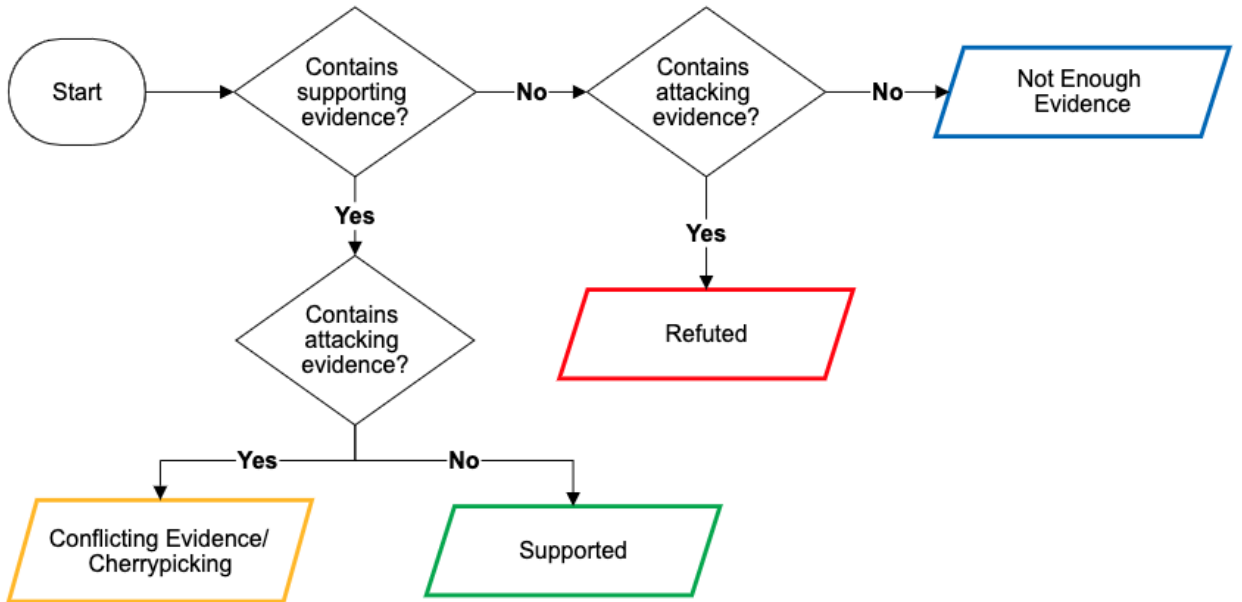


Figure 21: Flowchart depicting the algorithm for determining label predictions from evidence stances.

7.5.2 Argumentative method

Specific answers retrieved for questions were incoherent or irrelevant to the claim; they were filtered out using the spaCy model⁶ with word vectors, determining if the similarity between an answer and

⁶<https://spacy.io/usage/spacy-101>

its question surpassed a threshold of 0.3. The value of 0.3 was reached by comparing known relevant and irrelevant question-answer pairs to ensure the correct output was given in each case.

Evidence source URLs were distilled to their root domains and compared to reference lists of websites with high, medium and low credibility. Table 14 gives examples of source URLs, their root domains processed using the `tlldextract` module⁷, and their credibility retrieved by cross-referencing domains with the reference lists and explanations for the credibility.

Original URL	Root Domain	Credibility	Explanation
https://www.dailymail.co.uk/news/article-2006317/Sean-Connery-letter-Steve-Jobs-rejecting-offer-appear-Apple-ad-revealed-fake.html	www.dailymail.co.uk	Low	It is a tabloid newspaper with sensationalism and lack of fact-checking.
https://en.wikipedia.org/wiki/Billie_Eilish	en.wikipedia.org	Medium	Articles are combined using cited references but are open to editing from the public (subject to moderation).
https://www.congress.gov/congressional-record	www.congress.gov	High	It is an official government-owned website.

Table 14: Examples of evidence source URLs, their credibility and explanations.

Arguments were formed from question-answer pairs, with the answer as the premise and a conclusion created based on the stance of the evidence. An argument a_i can be expressed as a pair consisting of one premise p and one conclusion c : $a_i = (p_i, c_i)$. Table 15 shows an example of arguments in an argumentative explanation for a veracity prediction on a claim. This framework enforces that all conclusions must imply a prediction based on the evidence stance. Conclusions which imply the same prediction as the label prediction are labelled with $\hat{y}$. For example, for a claim with a prediction of ‘Not Enough Evidence’, a conclusion with a neutral stance will be labelled as $\hat{y}$.

Claim: Donald Trump delivered the largest tax cuts in American history. Label: Refuted		
Evidence	Stance	Argument
Question: Did the 2017 tax bill deliver the largest tax cuts in American history? Answer: This tax cut is the 8th largest as a percent of Gross Domestic Product (GDP) since 1918 and the 4th largest in inflation-adjusted dollars.	Attack	a_1 : This tax cut is the 8th largest as a percent of Gross Domestic Product (GDP) since 1918 and the 4th largest in inflation-adjusted dollars (p_1). This evidence refutes the claim ($c_1 = \hat{y}$).
Question: Has there been larger tax bills than the 2017 tax bill? Answer: Three tax bills have been larger: American Taxpayer Relief Act of 2012 (enacted in 2013); Tax Relief, Unemployment Insurance Reauthorization, and Job Creation Act of 2010 and Recovery Tax Act of 1981.	Attack	a_2 : Three tax bills have been larger: American Taxpayer Relief Act of 2012 (enacted in 2013); Relief, Unemployment Insurance Reauthorization, and Job Creation Act of 2010 and Recovery Tax Act of 1981 (p_2). This evidence refutes the claim ($c_2 = \hat{y}$).

Table 15: Examples of arguments in an argumentative explanation.

Stance aggregation

The Discontinuity-Free Quantitative Argumentation Debates (DF-QuAD) algorithm [79] considers both supporting and opposing arguments and determines the credibility of an argument based on its dialectical strength, ie. how many arguments support or attack that one [80]. The stance aggregation method employed in the VeriVic system expands this by determining the credibility of an argument by evaluating the trustworthiness of the evidence source while also balancing the number of arguments which support or attack the argument.

⁷<https://pypi.org/project/tldextract/>

In this paper, dialectical strength is defined by the credibility of the evidence source multiplied by the strength of the support-attack relations, where positive values indicate stronger agreement with the other arguments, while negative values indicate many attacks on this argument. The dialectical strength of argument a_i , D_{a_i} , is given by equation 12.

$$D_{a_i} = \nu_c \times (\nu_{Sup_{a_i}} - \nu_{Att_{a_i}}) \quad (12)$$

where $\nu_c \in [0, 1]$ is the credibility score of the argument's evidence source and $\nu_{Sup_{a_i}}$, $\nu_{Att_{a_i}}$ are defined in equations 13 and 14.

$$\nu_{Sup_{a_i}} = \sum_{a_j \in A} \mathbb{K}_{(a_j, a_i) \in R_{Sup}} \quad (13)$$

where A is the set of all arguments, R_{Sup} is the set of support relations, $(a_j, a_i) \in R_{Sup}$ means that a_j supports a_i . $\mathbb{K}_{(a_j, a_i) \in R_{Sup}}$ is an indicator function that equals 1 if $(a_j, a_i) \in R_{Sup}$, and 0 otherwise.

$$\nu_{Att_{a_i}} = \sum_{a_j \in A} \mathbb{K}_{(a_j, a_i) \in R_{Att}} \quad (14)$$

where A is the set of all arguments, R_{Att} is the set of attack relations, $(a_j, a_i) \in R_{Att}$ means that a_j attacks a_i . $\mathbb{K}_{(a_j, a_i) \in R_{Att}}$ is an indicator function that equals 1 if $(a_j, a_i) \in R_{Att}$, and 0 otherwise.

Table 16 and figure 22 demonstrate an example of arguments, support relations and attack relations in an Argumentative Explanation. Since Conflicting Evidence/Cherrypicking claims have both supporting and attacking evidence, all arguments with support or attack stance contribute to the prediction and thus have a conclusion $c_i = \hat{y}$.

Claim: Margaret Sanger was a racist who believed in eugenics. Her goal when founding Planned Parenthood was to eradicate minorities. Label: Conflicting Evidence/Cherrypicking			
Argument	Stance	Support relations	Attack relations
a_1 : Eugenics is the practice or advocacy of improving the human species by selectively mating people with specific desirable hereditary traits. It aims to reduce human suffering by ‘breeding out’ disease, disabilities and so-called undesirable characteristics from the human population (p_1). There is not enough evidence to support or refute the claim (c_1).	Neutral	None	None
a_2 : I think the greatest sin in the world is bringing children into the world – that have disease from their parents, that have no chance in the world to be a human being practically. Delinquents, prisoners, all sorts of things just marked when they’re born. That to me is the greatest sin – that people can – can commit (p_2). This evidence supports the claim ($c_2 = \hat{y}$).	Support	(a_2, a_4) , (a_4, a_2)	(a_2, a_3) , (a_3, a_2)
a_3 : As her biographer Ellen Chesler told me, she was a progressive who believed in racial integration. She voted for Norman Thomas. She worked with progressive Black people - W.E.B. Du Bois, for example, who along with Mary McCleod Bethune and Adam Clayton Powell Sr. served on the board of the Negro Project, a network of birth control and maternal health clinics Sanger established in Harlem and the South. In 1966, Martin Luther King accepted Planned Parenthood’s first Margaret Sanger Award, and in his statement offered a vigorous endorsement of voluntary birth control (p_3). This evidence refutes the claim ($c_3 = \hat{y}$).	Attack	None	(a_2, a_3) , (a_3, a_2) , (a_4, a_3) , (a_3, a_4)
a_4 :Although she did not single out Black people, Sanger was, yes, a eugenicist. She thought people, especially poor people, often had too many kids to care for properly and that too many of those kids were born physically disabled (or in the language of the day, ‘feeble-minded’). She did not oppose forced sterilization (p_4). This evidence supports the claim ($c_4 = \hat{y}$)	Support	(a_4, a_2) , (a_2, a_4)	(a_4, a_3) , (a_3, a_4)

Table 16: Examples of arguments in an argumentative explanation.

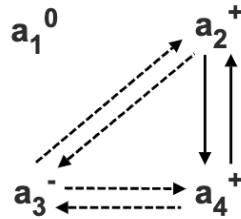


Figure 22: Diagram of arguments in an argumentative explanation including support and attack relations. Solid lines represent bipolar support relations and dotted lines represent bipolar attack relations between arguments. For arguments, + indicates a support stance, - is an attack stance and 0 is a neutral stance.

The `determine_veracity_with_explanation` aggregates the scores for each argument stance and determines whether the claim is supported, refuted, conflicting, or has not enough evidence. If the top scores between support and attack stances are close (within 0.1), it predicts Conflicting Evidence/Cherrypicking, otherwise it favours the stronger stance. If the top stance is ‘attack’, the prediction is Refuted, similarly for a support stance and Supported. Only if all stances are neutral will the prediction be Not Enough Evidence.

7.5.3 Analysis

Rules-based Method

Effect of Different Models

Model 1, which lacked text pre-processing and had questions in addition to answers as the evidence context, produced the highest F1 score for Refuted claims. Model 4, which was trained on less data than all others with a train/test split of 0.8/0.2 instead of 0.9/0.1, performed the worst on ‘Not Enough Evidence’ claims. It was hypothesised that since the 500-claim dataset was randomly shuffled before the train-test split, perhaps there was a lower percentage of these claims in the 0.8 split compared to the 0.9 split. The reduced representation in the training data could explain the reduced performance since there was less data in this class for model 4 to fit to. However, figures 23 and 24 show that there is a negligible reduction of 0.2%. Model 5 performed the worst on the Conflicting Evidence/Cherrypicking class. With the additions to model 5 of text pre-processing and answer-only context in model 6, there was a stark improvement. Table 17 shows that the F1 Conflicting Evidence/Cherrypicking increased from 0.066 to 0.182 between models 5 and 6.

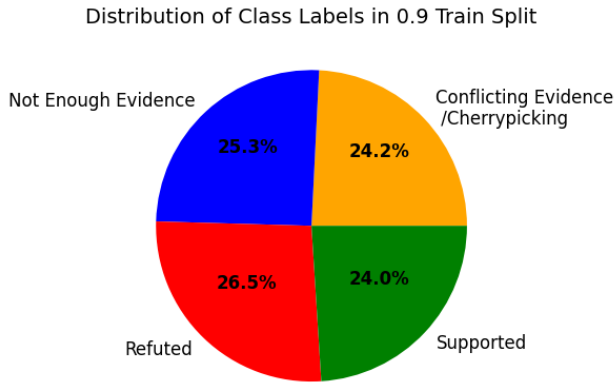


Figure 23: Distribution of class in the 0.9 training split of the training data.

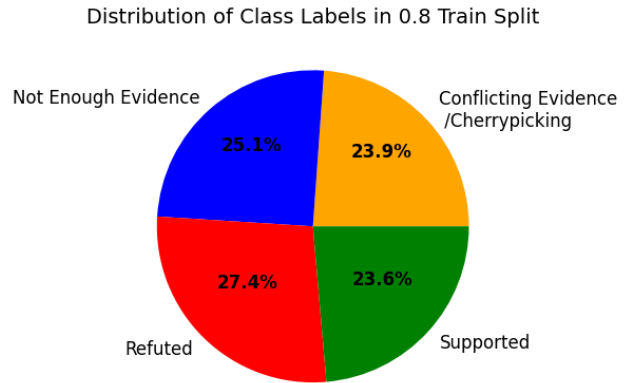


Figure 24: Distribution of class in the 0.8 training split of the training data.

Effect of Class Abundance in Dataset

Figures 25 and 26 show larger class representation in the dataset corresponded to better F1 score performance while the two smallest classes suffered with poor performance. This trend is also present in the AVeriTeC Baseline results summarised at the bottom of tables 17 and 18 where the the F1 scores per class (and abundance in the development dataset) are as follows: F1 Refuted = 0.69 (61%), F1 Supported = 0.41 (24%), F1 Not Enough Evidence = 0.16 (7.6%) and F1 Conflicting Evidence/Cherrypicking = 0.10 (7.0%). Errors in predictions for the ‘Refuted’ (61% abundance in development dataset) class have a lesser effect on the F1 score than that of smaller classes such as ‘Not Enough Evidence’ (7.6% abundance) and ‘Conflicting Evidence/Cherrypicking’ (7.0% abundance).

Experiment	F1 Sup	F1 Ruf	F1 Not	F1 Con	Macro F1	AVeriTeC Score
1	0.286	0.611	0.203	0.162	0.315	0.112
2	0.268	0.564	0.186	0.133	0.288	0.092
3	0.259	0.562	0.213	0.104	0.285	0.108
4	0.360	0.508	0.136	0.129	0.283	0.110
5	0.363	0.473	0.200	0.066	0.275	0.110
6	0.220	0.590	0.192	0.182	0.296	0.106
AVeriTeC Baseline	0.410	0.690	0.160	0.100	0.230	0.150

Table 17: F1 Scores and AVeriTeC scores for experiments conducted on dataset with Q-only HU-METEOR of 0.38 and QnA HU-METEOR of 0.20.

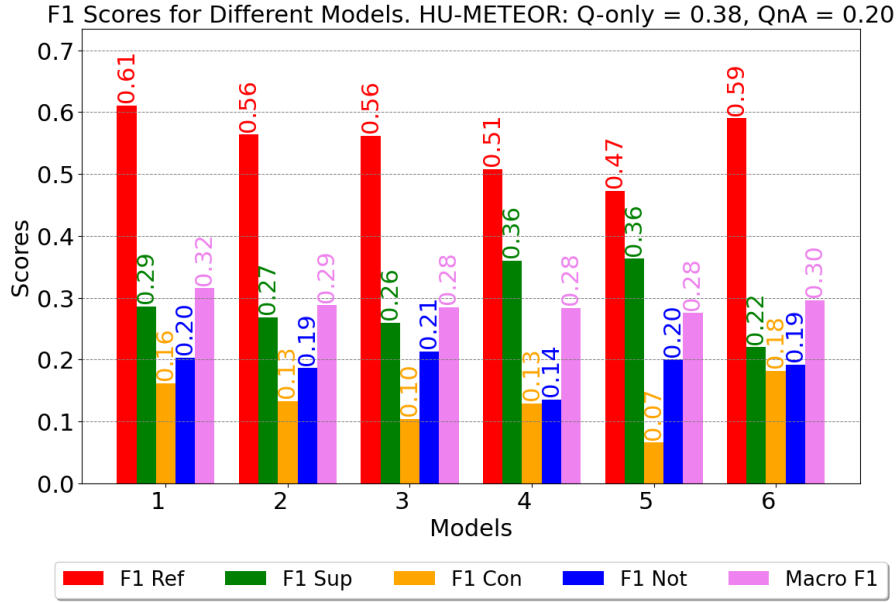


Figure 25: F1 scores for different models for development claims with Q-only HU-METEOR of 0.38, an average of 2.99 questions.

Experiment	F1 Sup	F1 Ruf	F1 Not	F1 Con	Macro F1	AVeriTeC Score
1	0.280	0.610	0.144	0.119	0.288	0.132
2	0.327	0.570	0.169	0.122	0.297	0.126
3	0.284	0.565	0.168	0.121	0.285	0.142
4	0.393	0.532	0.093	0.176	0.299	0.126
5	0.455	0.430	0.136	0.086	0.277	0.142
6	0.319	0.605	0.151	0.113	0.297	0.142
AVeriTeC Baseline	0.410	0.690	0.160	0.100	0.230	0.150

Table 18: F1 Scores and AVeriTeC scores for experiments conducted on dataset with Q-only HU-METEOR of 0.40 and QnA HU-METEOR of 0.22.

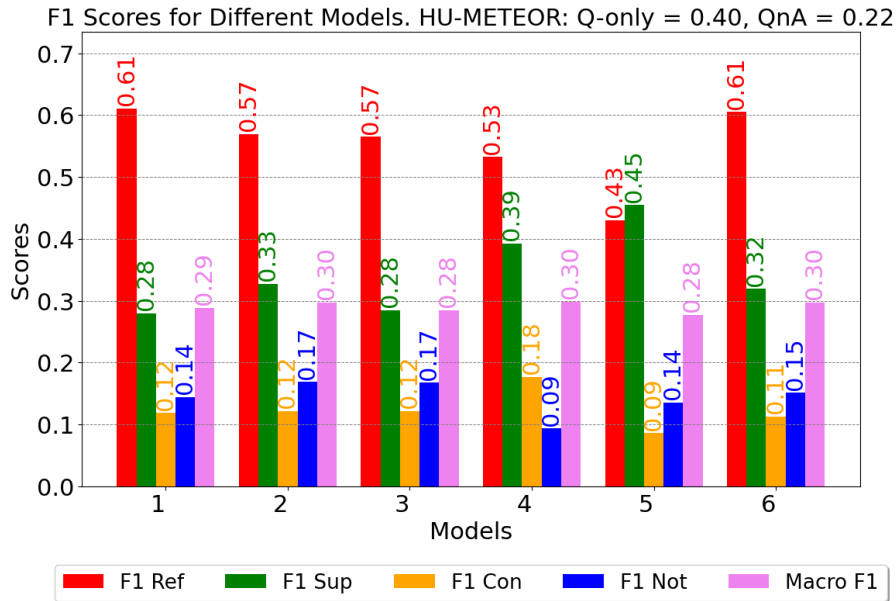


Figure 26: F1 scores for different models for development claims with Q-only HU-METEOR of 0.40, average of 3.37 questions.

Effect of Average Number of Questions

Figure 27 shows that evidence with a higher HU-METEOR score for questions and answers had a higher AVeriTeC score, evident across all models. It was discussed earlier that a higher HU-METEOR was achieved with a higher number of questions generated. Therefore, I can conclude that increasing the number of questions (and subsequently answers) generated improves the AVeriTeC score. It must be noted that none of the results surpassed the baseline result of 0.15 which suggested that more questions were required.

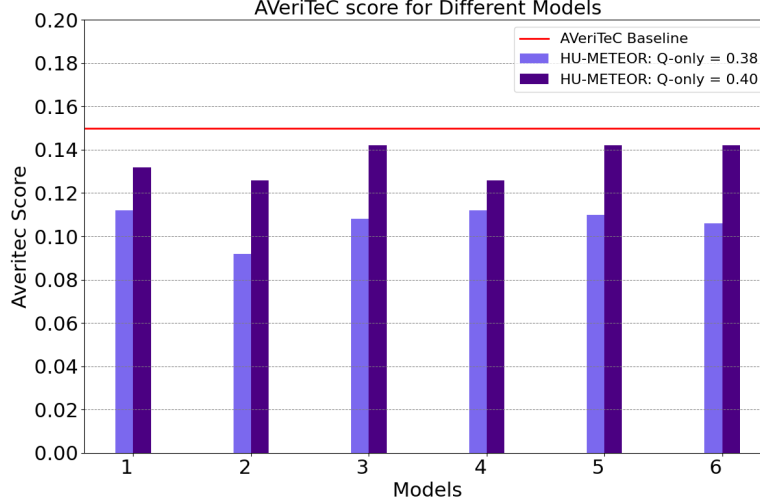


Figure 27: AVeriTeC scores for different models. The datasets with a Q-only HU-METEOR of 0.38 and 0.40 have averages of 2.99 and 3.37 questions respectively.

Effect of Answer Retrieval

Table 18 revealed that for a higher number of generated questions, the RoBERTa models achieved higher AVeriTeC scores than the BERT model. For further experimentation, data with an average of 6.38 questions per claim and a Q-only HU-METEOR of 0.46 was used to investigate the different types of answer retrieval methods.

Figures 28 and 29 show that performance is comparable across answer retrieval types but the RoBERTa-based model 6 has better F1 score performance in every F1 score metric. Table 19 shows that the dataset with answer generated by T5 had the highest F1 across 75% of the classes in addition to the highest macro F1 score for model 6. While the Initial answer type dataset achieved the best AVeriTeC score with model 6 (see table 30), I determined that the best result was given running inference on the T5 data answer type data. This is because while the AVeriTeC scores are identical to 2 significant figures (0.16), the T5 data achieved the best macro F1 performance.

Answer Type	Model	F1 Sup	F1 Ruf	F1 Not	F1 Con	Macro F1	AVeriTeC Score
T5	3	0.239	0.510	0.183	0.120	0.263	0.146
T5	6	0.313	0.575	0.242	0.139	0.318	0.158
RoBERTa-base-SQuAD2	3	0.169	0.515	0.178	0.121	0.245	0.126
RoBERTa-base-SQuAD2	6	0.291	0.586	0.220	0.136	0.308	0.160
Initial	3	0.182	0.510	0.146	0.131	0.242	0.142
Initial	6	0.313	0.574	0.195	0.126	0.302	0.164
AVeriTeC Baseline		0.410	0.690	0.160	0.100	0.230	0.150

Table 19: Effect of answer retrieval type on F1 and AVeriTeC score for models 3 (BERT-large-uncased with pre-processing and answer-only context) and 6 (RoBERTa-large with pre-processing and answer-only context).

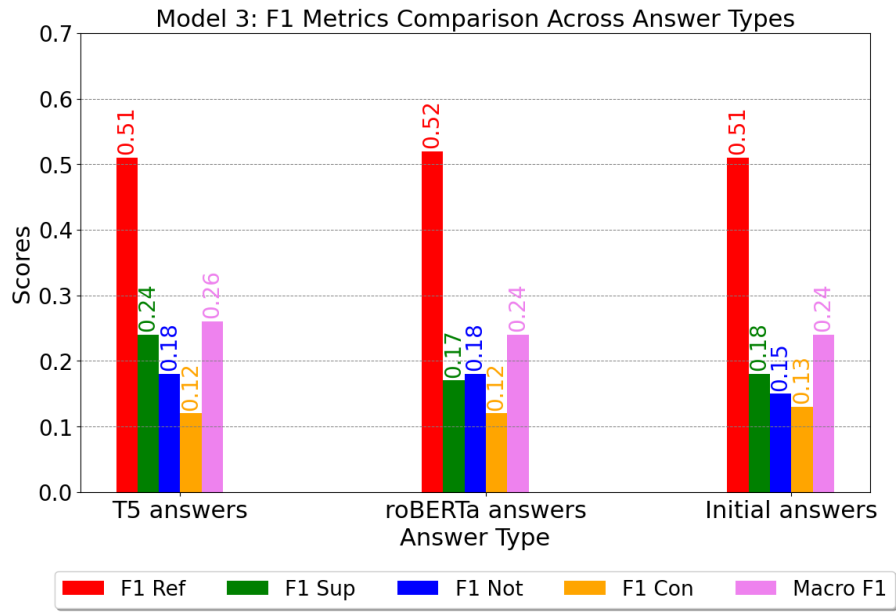


Figure 28: F1 scores for different answer retrieval types using Model 3.

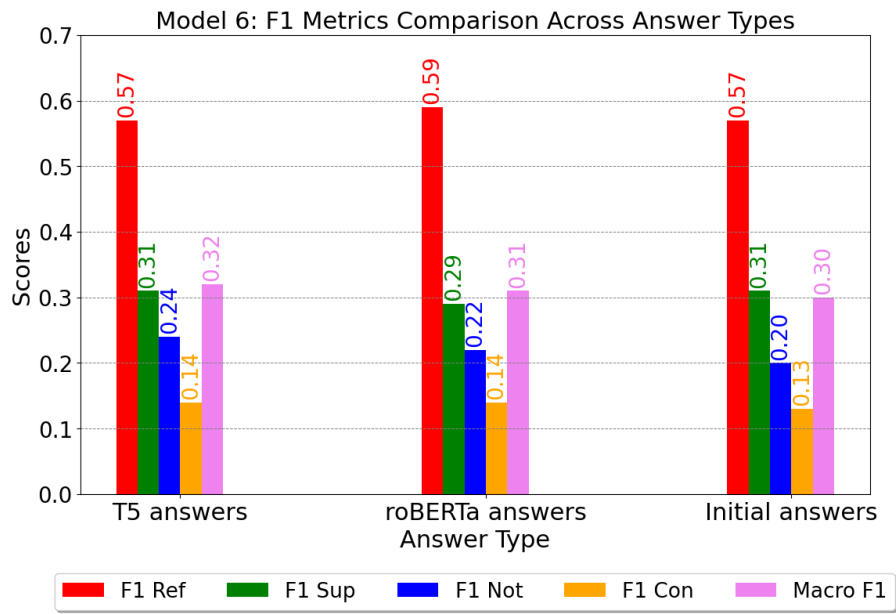


Figure 29: F1 scores for different answer retrieval types using Model 6.

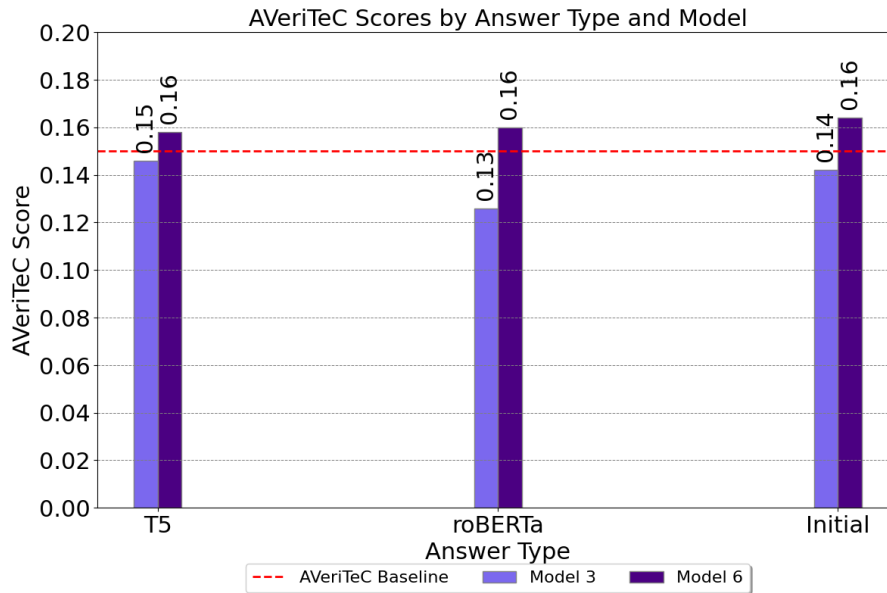


Figure 30: AVeriTeC scores for answer retrieval types. Model 3 is a BERT-large-uncased model finetuned with preprocessed claims and answers from the 500-claim stance detection training set. Model 6 is a RoBERTa-large model finetuned in the same way.

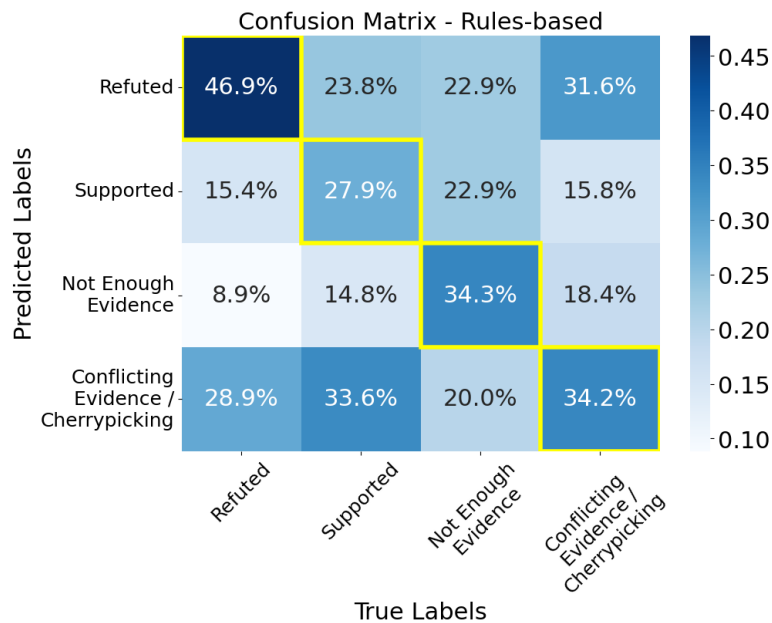


Figure 31: Confusion matrix for label predictions generated using a Rules-based method.

Argumentative Method

Figure 33 shows that using the Argumentative framework to generate veracity predictions achieved the highest Macro F1 score and highest F1 score for the Refuted and Conflicting Evidence/Cherrypicking classes. Despite this, the Supported class' F1 score decreased by 39% when compared with the AVeriTeC-BLOOM-7b system, this is because the majority (58.2%, figure 32) of Supported claims were mispredicted as Refuted.

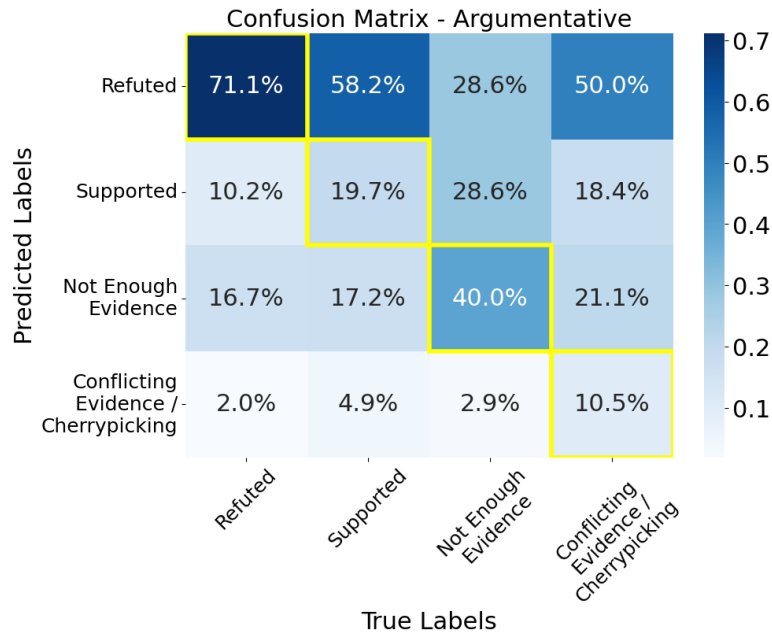


Figure 32: Confusion matrix for label predictions generated using a BAF.

The Argumentative method reduced the number of Supported and Refuted claims that were misclassified as Conflicting Evidence/Cherry-picking, evidenced by figures 31 and 32. Instead of applying this label to any claim with supporting and attacking evidence, the difference between the dialectical strength of the opposing stance had to be less than 0.1. The number of Refuted and Supported claims misclassified as Conflicting Evidence/Cherry-picking decreased from 28.9% and 33.6% to 3.0% and 4.9%, respectively, by employing the Argumentative method.

The Argumentative method achieved the highest AVeriTeC score with a 53% increase on the AVeriTeC-BLOOM-7b system (figure 34). This is due to the high accuracy (71.1%) for the Refuted class which constitutes the majority (61.0%) of the development dataset.

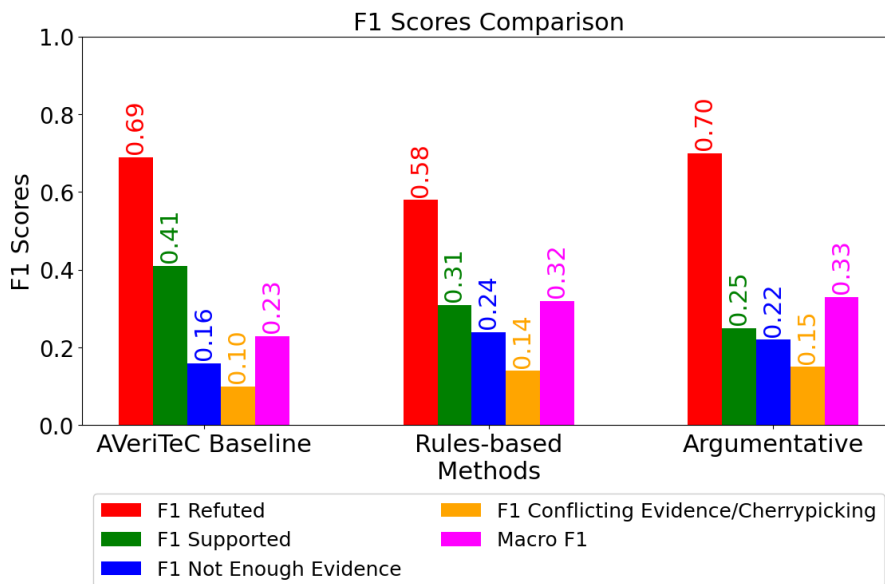


Figure 33: F1 scores for predictions made using a BAF.

7.6 Justification generation

The justifications for both the Rules-based and Argumentative predictions were template-based and inspired by the gold-justifications. For Supported and Refuted predictions, sections of the claim were included as seen in tables 20 and 21. While the Rules-based templates included the specific evidence which were responsible for the prediction, the Argumentative method vaguely describe the stances of the arguments to justify the prediction.

Prediction	Justification template
Refuted	The evidence shows that it false that {claim}. Therefore the claim is refuted.
Supported	The evidence shows that {claim}. Therefore the claim is supported.
Not Enough Evidence	QA pairs {list of indices for evidence with no answers} are unanswerable. QA pairs {list of neutral evidence indices} is neutral to the claim. Therefore there is not enough evidence to refute or support the claim.
Conflicting Evidence /Cherrypicking	Although QA pairs list of indices for supporting evidence support the claim, QA pair {list of indices for refuting evidence} attack the claim. Therefore the claim has conflicting evidence or is cherry picking.

Table 20: Prediction justification templates based for the Rules-based method.

Prediction	Justification template
Refuted	The evidence shows that it false that {claim}. Therefore the claim is refuted.
Supported	The evidence shows that {claim}. Therefore the claim is supported.
Not Enough Evidence	All arguments are neutral towards the claim. Therefore there is not enough evidence to refute or support the claim.
Conflicting Evidence /Cherrypicking	Some arguments support the claim but some arguments attack the claim. Therefore the claim has conflicting evidence or is cherry picking.

Table 21: Prediction justification templates based for the Argumentative method.

7.6.1 Analysis

Figure 34 shows that the Argumentative method yielded the highest Justification METEOR Score (0.12) compared to the AVeriTeC-BLOOM-7b system (0.07) and Rules-based method (0.11). Although the Averitec Score for the Argumentative method was 44% greater than that of the Rules-based method, the Justification METEOR Score was only 9% greater. This is because while the general format of the template-based Argumentative justifications was similar to the gold justifications, the common abbreviation “QA” (question and answer) was missing, reducing the similarity.

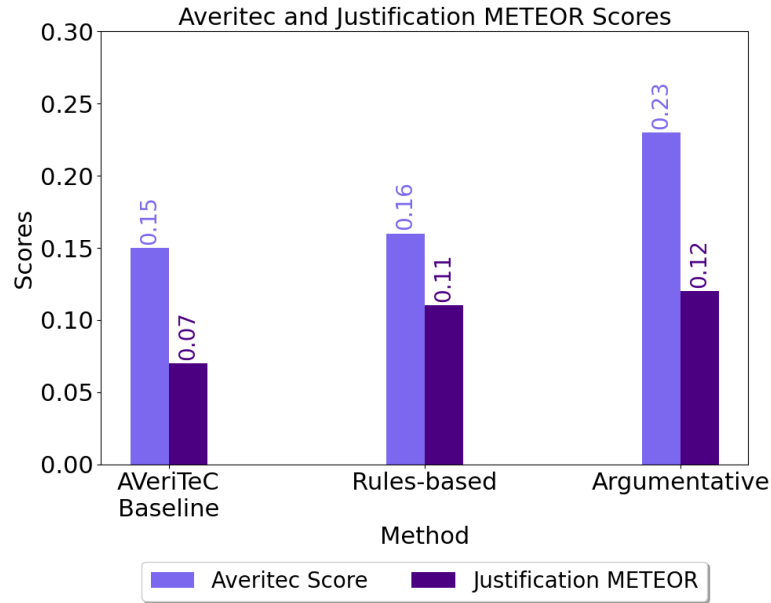


Figure 34: Averitec Scores and Justification METEOR Scores for AVeriTeC-BLOOM-7b, the Rules-based method and the Argumentative method.

Figure 35 show that the arguments in the Not Enough Evidence and Conflicting Evidence/Cherry picking had perfect acceptability, indicating that each argument with the conclusion $\hat{y}$ did not have any attacks. Supported claims had the lowest average acceptability of 0.86 due to arguments with support conclusions ($c_i = \hat{y}$) being attacked by arguments with attack conclusions. To improve the acceptability, I propose incorporate the attacking argument’s dialectical strength with the outlook to weaken attacking arguments if they have low credibility or are attacked by many other arguments.

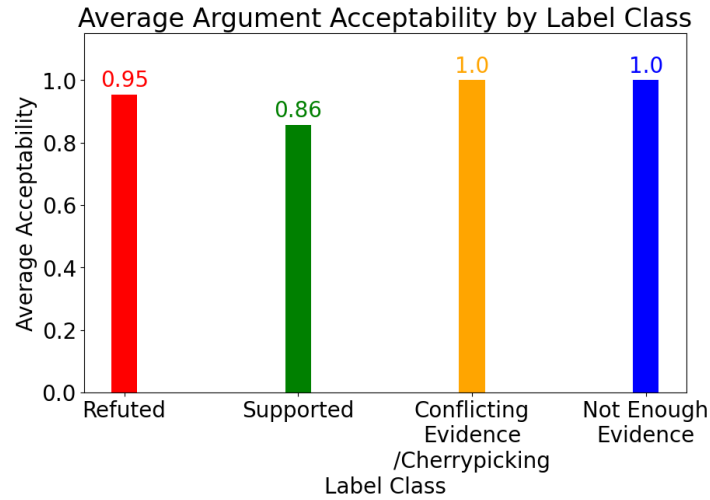


Figure 35: Average dialectical acceptability scores for each label class. Perfect acceptability is indicated by a value of 1.0.

Only arguments for Not Enough Evidence claims had perfect non-circularity, since their neutral stances lead to zero attack or support relations. The design of the argumentation framework used in the VeriVic system caused high levels of dialectical circularity between the arguments of the other label classes. Figure 22 shows that if argument a_i attacks argument a_j , then a_j also attacks argument a_i because support and attack relations were determined by two arguments having the the same or different stances. To improve the non-circularity, I propose creating a hierarchy of arguments based on publishing date as demonstrated in figure 37. In this way, the argument based on the earliest evidence

would be the base argument with later claims attacking or supporting them. It allows for one-way support and attack relations to reduce circularity.

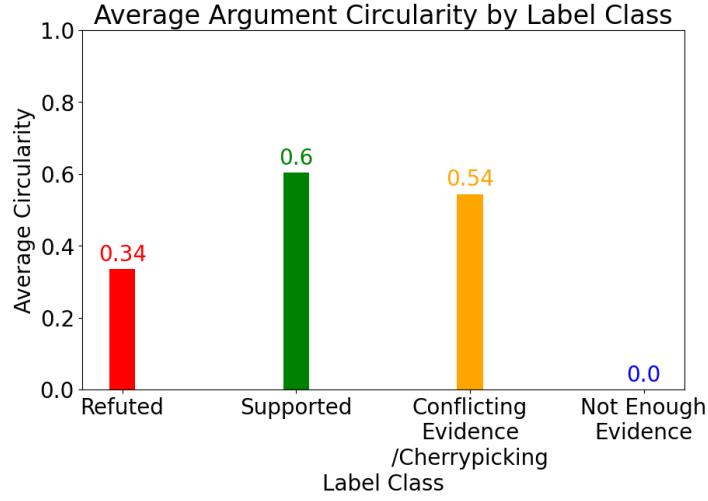


Figure 36: Average dialectical circularity of arguments for each label class. Perfect non-circularity is indicated by a value of 0.0.

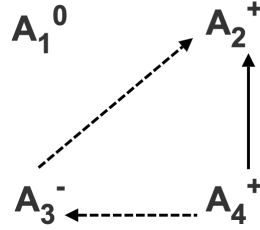


Figure 37: Non-circular arguments in a hierarchy of evidence publishing date. From earliest to latest publishing date: $a_1^0, a_2^+, a_3^-, a_4^+$. Solid lines represent binary support relations and dotted lines represent binary attack relations.

8 VeriVic System Summary

8.1 System Description Summary

Question Generation

Questions were generated using the BART-base model in two ways: (1) 10 questions directly from the claim (DIR method), (2) 10 questions from predicates derived from the claim (FOL method). The dataset for method 1 was created by extracting claims and their questions from the AVeriTeC training set. The FOL datasets were created by prompting GPT-4o to create predicates from gold questions; the first dataset consisted of claim-predicate pairs, and the second consisted of predicate-question pairs. The questions from the DIR and FOL methods were combined and filtered to ensure uniqueness and reduce unnecessary evidence duplicates.

Search and Evidence Retrieval

The generated questions were used as Google Search API and DuckDuckGo API queries to retrieve a maximum of 10 evidence documents published before the claim date. Claim metadata was used in cases where necessary in line. The Beautiful Soup library was used to extract sentences from the evidence documents and rank them according to similarity to the respective query. Any remaining unanswered questions were processed by the T5 model for question answering.

Stance Detection, Veracity Prediction and Justification Generation

RoBERTa-large was used to determine each question-answer evidence pair’s stance (support, attack,

neutral). Evidence pairs were converted into arguments consisting of a premise (answer) and a conclusion (stance). Credibility was determined according to the trustworthiness of the evidence source, and support and attack relations between arguments were aggregated together to determine the label predictions. Explanations were generated using templates according to the stance aggregation.

8.2 Analysis

Table 22 and figure 38 show that the VeriVic system outperforms the AVeriTeC-BLOOM-7b in every metric. Using an Argumentative explanation framework improved the quality of the justifications by 71%. Improving the F1 score for Conflicting Evidence/Cherrypicking claims by 50% means the VeriVic system is better at identifying and correctly categorising claims with contradictory or selective evidence than AVeriTeC-BLOOM-7b. Since these claims are a significant challenge in real-world fact-checking, improving performance in this area lead to the improvement of the Averitec Score. This improvement reflects a system that can more accurately navigate the complex landscape of claim verification, especially in contentious topics where opposing information is presented. AVeriTeC annotators acknowledged that there was bias during the evidence annotation stage which meant that certain Refuted claims did have supportive evidence which was not included [15]. Therefore stance aggregation as described in this paper is vital to be able to distinguish between high quality and lower quality evidence to ensure that the opposing evidence does not lead to mispredictions.

While the Question-only HU-METEOR Score increased by 77%, the Question and Answer HU-METEOR Score only grew by 19%. This lack of proportionality is due to the fact that the AVeriTeC-BLOOM-7b system had access to a knowledge store which contained gold evidence documents amongst others, increasing the chances that gold evidence (or very similar) would be retrieved. However, I opted to not limit my web search to a closed list of websites as real-world scenarios allow for free searching of the internet to obtain evidence. I hypothesise that using this knowledge store as the list of evidence documents would further improve the Q+A HU-METEOR Score.

System	Q-only	Q+A	Averitec	F1 Ref	F1 Sup	F1 Not	F1 Con	Macro F1	Justification
AVeriTeC-BLOOM-7b	0.26	0.21	0.15	0.69	0.41	0.16	0.10	0.23	0.07
VeriVic	0.46	0.25	0.23	0.70	0.25	0.22	0.15	0.33	0.12

Table 22: Comparison of evaluation metrics for AVeriTeC-BLOOM-7b and VeriVic systems.

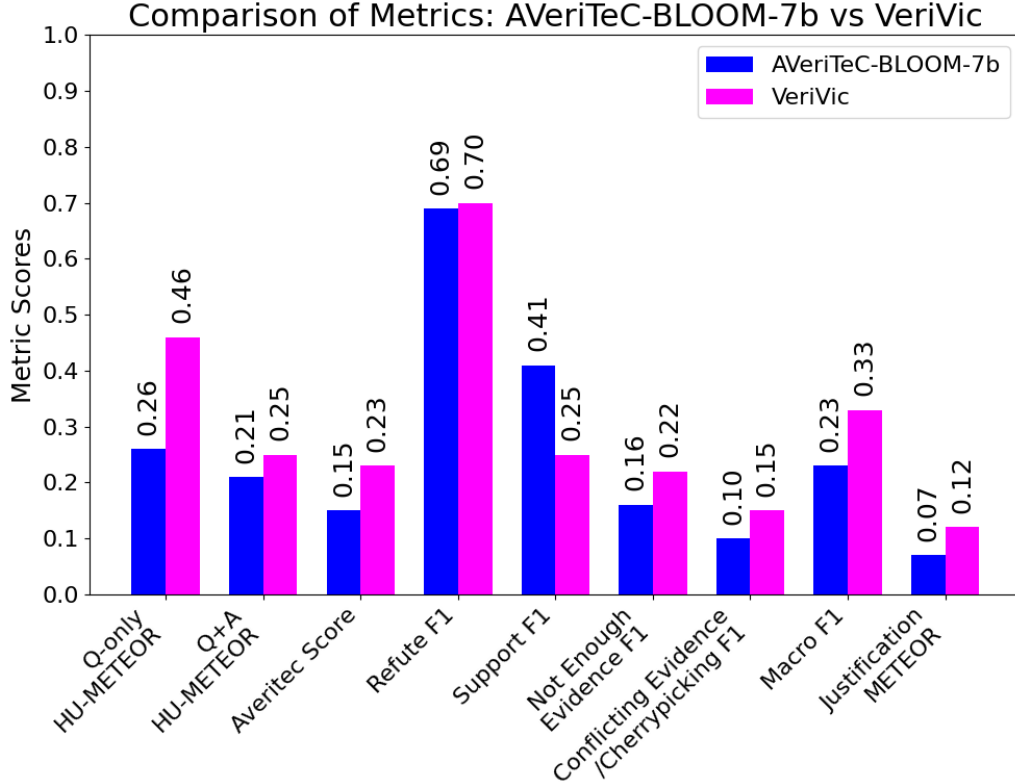


Figure 38: Comparison of evaluation metrics for AVeriTeC-BLOOM-7b and VeriVic systems.

9 Conclusion

This research detailed throughout this paper brought about the VeriVic System, an Automated Fact-Checking System developed to predict the veracity of real-world claims and provide explainability by generating coherent prediction justifications. The main aims were to develop a straightforward approach to evidence retrieval, improve the performance for predictions on the novel ‘Conflicting Evidence/Cherrypicking’ claims and use Argumentative Explanations to improve the Averitec Score.

I simplified the evidence retrieval stage by using a custom ranking method to extract answer sentences from a list of only 10 evidence documents per question. To ensure maximum evidence retrieved, a T5 model was used for question answering, taking context from the most relevant evidence document per question. This achieved a Question and Answer HU-METEOR of 0.25, a 19% improvement on the AVeriTeC-BLOOM-7b system. The Bipolar Argumentative Framework used improved the accuracy of veracity prediction with an AVeriTeC score of 0.23 while improving the explainability of the process. This is evident in the increased Justification METEOR Score of 0.12 compared to AVeriTeC-BLOOM-7b’s 0.07. This method achieved the goal of improving the performance for the minority class, Conflicting Evidence/Cherrypicking, by 50%.

These findings are significant because it presents an easily interpretable method of claim verification which enhances the accuracy of detecting cherry-picked claims. This class of claims are very important to be able to distinguish as they may technically be true but are misleading, posing a covert threat of misinformation. Due to the open-domain nature of this system, with its access to the entire internet, it allows for wide applications within automated fact checking.

10 Future Work

Despite significant breakthroughs and interesting findings in this project, there are many areas for future improvement. If more resources were available, there is ample opportunity to finetune the

state-of-the-art Large Language Models from OpenAI for various tasks. Additionally, if there was more time available, detailed hyperparameter tuning could be employed to optimise the models used.

During the Evidence retrieval, PDF files and websites which required cookie acceptance caused timeout errors and were not accessed. In the future, work could be done to circumvent these issues using the requests library for example.

The evidence retrieval score (Q+A HU-METEOR) could be improved if the AVeriTeC knowledge store was used instead of web search. This knowledge store contained lists of relevant webpages for a set of predetermined questions; some web pages contained the gold evidence.

The argumentative explanations used had high dialectical circularity since arguments in bipolar support or attack relations supported or attacked each other. To improve, I suggest ordering the arguments based on the chronology of evidence sources where the argument based on the earliest evidence source would be the first argument in the explanation. In this way, there would be one-way support or attack relations. More work could be done to the Argumentative explanations' justification templates to structure it in a more debative style.

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A Appendix A

The task is to:

- 1) define all the predicates in the claim
- 2) parse the predicates into follow up questions

Claim: Howard University Hospital and Providence Hospital are both located in Washington, D.C.

Predicates:

1. Location(Howard Hospital, Washington D.C.) - Verify Howard University Hospital is located in Washington, D.C.
2. Location(Providence Hospital, Washington D.C.) - Verify Providence Hospital is located in Washington, D.C.

Follow up Questions:

1. Where is Howard Hospital located?
2. Where is Providence Hospital located?

Claim: An IndyCar race driver drove a Formula 1 car designed by Peter McCool during the 2007 Formula One season.

Predicates:

1. Designed(Peter McCool, a Formula 1 car) - Verify a Formula 1 car was designed by Peter McCool during the 2007 Formula One season.
2. Drive(An IndyCar race driver, a Formula 1 car) - Verify an IndyCar driver drove a Formula 1 car.

Follow up Questions:

1. Which Formula 1 car was designed by Peter McCool during the 2007 Formula One season?
2. Did an IndyCar driver drove a Formula 1 car designed by Peter McCool during the 2007 Formula One season?

Claim: Thomas Loren Friedman has won more Pulitzer Prizes than Colson Whitehead

Predicates:

1. Won(Thomas Loren Friedman, Pulitzer Prize) - Verify the number of Pulitzer Prizes Thomas Loren Friedman has won.
2. Won(Colson Whitehead, Pulitzer Prize) - Verify the number of Pulitzer Prizes Colson Whitehead has won.

Follow up Questions:

1. How many Pulitzer Prize did Thomas Loren Friedman win?
2. How many Pulitzer Prize did Colson Whitehead win?

Claim: SkyHigh Mount Dandenong (formerly Mount Dandenong Observatory) is a restaurant located on top of Mount Dandenong, Victoria, Australia.

Predicates:

1. Location(SkyHigh Mount Dandenong, top of Mount Dandenong, Victoria, Australia) ::: Verify that SkyHigh Mount Dandenong is located on top of Mount Dandenong, Victoria, Australia.
2. Known(SkyHigh Mount Dandenong, Mount Dandenong Observatory) ::: Verify that SkyHigh Mount Dandenong is formerly known as Mount Dandenong Observatory.

Follow up Questions:

1. Where is SkyHigh Mount Dandenong located?
2. Was SkyHigh Mount Dandenong formerly known as Mount Dandenong Observatory?

Claim: Donald Trump delivered the largest tax cuts in American history.

Predicates:

Follow up questions:

Figure 39: Prompt given to GPT-4o using 4-shot learning for generating first-order logic predicates from claims and questions from the predicates.

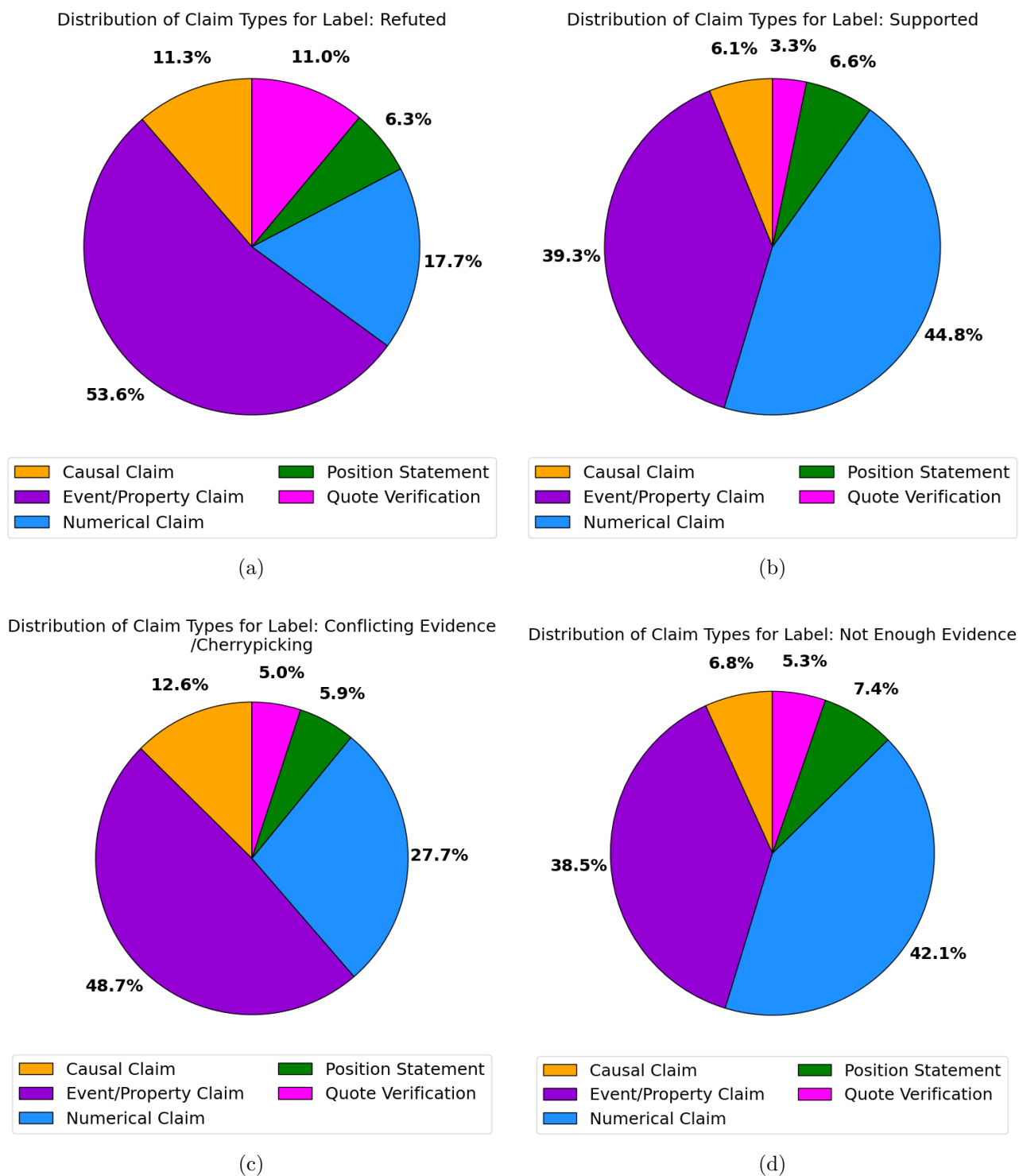


Figure 40: Claim types for per label class, a) Refuted, b) Supported, c) Conflicting Evidence/Cherry-picking, d) Not Enough Evidence.

Acceptability is formally defined by Kotonya and Toni [3] in equation 15.

$$Acc(\mathbf{E}) = \frac{1}{N} \sum_{a_i=(P,\hat{y}) \in A} \left(\frac{1}{|Atts(a_i)|} \sum_{(a_j, a_i) \in R_{Att}} \delta(a_j) \right) \quad (15)$$

where $Acc(\mathbf{E}) \in [0, 1]$, $Atts(a_i) = \{a_j | (a_j, a_i) \in R_{Att}\}$, $\delta(a_j) = 1$ if $(a_j, a_i) \in R_{Att}$ and $\delta(a_j) = 0$ otherwise [3].

Circular argumentation is formally defined in equation 16.

$$Cir(\mathbf{E}) = \frac{1}{N} \sum_{a \in A} \frac{1}{M} \sum_{A' \in R_{Sup}, R_{Att}} head(A', a) \cdot tail(A', a) \quad (16)$$